

# Elliptic Curves

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# 0 Motivation

**Definition 0.1** (Diophantine equation). Polynomial equation to be solved with integers.

**Examples.**

- (1) Pythagorean triples. To find  $(a, b, c) \in \mathbb{Z}^3$  such that  $a^2 + b^2 = c^2$ .
- (2) Fermat's Last Theorem. Fix  $n > 3$ , then  $x^n + y^n = z^n$  has only trivial solution in  $\mathbb{Z}$ .

Both of these examples are homogeneous, i.e. all monomials have the same degree. In general, if  $f(x_1, \dots, x_m) = 0$  with  $f$  homogeneous, then  $(\lambda x_1, \dots, \lambda x_m)$  is another solution.

Questions:

1. Do solutions exist?
2. How many?
3. Structured? Random?
4. Algorithms?

We would like to solve solutions over  $\mathbb{Q}$  and  $\mathbb{Z}$  and often  $\mathbb{Q}$  since it's a field.

**Examples** (Complexity of Polynomial Equations).

1. (Number of Variables) 1-variable  $\Leftrightarrow$  finding roots of 1-variable polynomial over  $\mathbb{Q}$  by rational root theorem.

We are more interested in 2 variables.

2. (Degree) Largest degree of a monomial term like  $x^3 + y^2 + z$  has degree 3.

**Example** (Degree-1, 2-variables).  $ax + by = c$ ,  $a, b, c \in \mathbb{Q}$ . To find all rational points, set  $x = t \in \mathbb{Q}$ . Since for  $y$ ,  $y = \frac{c-at}{b} \in \mathbb{Q}$ . Thus, the set of  $\mathbb{Q}$ -points is  $\{(t, \frac{c-at}{b}) : t \in \mathbb{Q}\}$ . By Euclidean algorithm, we see that it has a solution if and only if  $\gcd(a, b) | c$ . Find a solution  $(x_0, y_0)$  via Euclidean algorithm, any other solution is

$$(x_0 + \frac{\text{lcm}(a, b)}{a}t, y_0 - \frac{\text{lcm}(a, b)}{b}t) : t \in \mathbb{Z}$$

We note that once we find one solution, it's easy to find all others.

**Example** (Degree-3, 2-variables, Elliptic Curves). One specific example is the Congruent Number Problem: which areas  $n \in \mathbb{Z}$  arise as areas of right triangles with rational side length. Note that 6 is congruent with side lengths  $\{3, 4, 5\}$ , 5 is congruent with side lengths  $\{3/2, 20/3, 41/6\}$ . In fact the set of congruent is

$$\{5, 6, 7, 13, 14, 15, 20, 21, 22, 23, 24, \dots\}$$

There's a bijection

$$\{(a, b, c) \in \mathbb{Q}^3 : a^2 + b^2 = c^2, \frac{ab}{2} = n\} \rightarrow \{(x, y) \in \mathbb{Q}^2 : y^2 = x^3 - n^2x, y \neq 0\}$$

$$(a, b, c) \mapsto (\frac{nb}{c-a}, \frac{2n^2}{c-a})$$

and conversely

$$(x, y) \mapsto (\frac{x^2 - n^2}{y}, \frac{2nx}{y}, \frac{x^2 + n^2}{y})$$

**Example** (Example of an elliptic curve). Cubic curve with at least 1 rational point and well-defined tangents.

**Example.** For the degree of elliptic curve greater than 4 we need the following theorem.

**Theorem 0.2** (Faltings theorem). For "generic"  $\deg \geq 4$  curves in 2 variables, we have finitely many rational solutions.

**Example** (Degree-2, 2-variables). Asking about rational points on conics.

**Definition 0.3.** A curve defined by polynomial equation  $f(x_1, \dots, x_m) = 0$  is called an algebraic curve. If  $f \in \mathbb{Q}[x_1, \dots, x_m]$ , then the curve is a rational curve.

**Example.** Rational Lines, Rational Conics.

**Lemma 0.4** (Geometric principle). Let  $C$  be a curve and  $L$  be a line,  $C \cap L$  in a rational point  $P_1$  and also at  $P_2$ . Then

- (1) if  $C, L$  rational, then  $P_2$  rational.
- (2) if  $P_2$  rational, then  $L$  rational line.

**Proof.** To find intersections, substitute the equation for the line e.g.  $ax + by = c$  into the equation of  $C$ ,  $x$ -coordinates of intersections are roots of a quadratic with  $\mathbb{Q}$  coefficients. We know the sum of roots are rational so if one root is rational then the other one is rational as well.  $\square$

**Example.** Parametrize all rational points on  $x^2 + y^2 = 1$ . Choose  $P = (-1, 0)$  as an initial rational point. Choose line of slope  $\lambda \in \mathbb{Q}$  then  $L_\lambda : y = \lambda(x + 1)$ .  $L_\lambda$  intersects circle when

$$x^2 + \lambda^2(x + 1)^2 = 1 \implies (1 + \lambda^2)x^2 + 2\lambda^2x + \lambda^2 - 1 = 0$$

Hence, the other root is  $\frac{1-\lambda^2}{1+\lambda^2}$ . The  $y$  coordinate is  $\lambda(\frac{1-\lambda^2}{1+\lambda^2} + 1) = \frac{2\lambda}{1+\lambda^2}$ . Thus, the set of rational points on  $x^2 + y^2 = 1$  is

$$\{(\frac{1-\lambda^2}{1+\lambda^2}, \frac{2\lambda}{1+\lambda^2}) : \lambda \in \mathbb{Q} \cup \{\infty\}\}$$

**Remark (Euclidean).** Pythagorean triple can be parametrized to

$$k \cdot (n^2 - m^2, 2mn, n^2 + m^2), m, n \in \mathbb{Z}, k \in \mathbb{Z}$$

where  $m, n$  has opposite parity and we can swap  $n^2 - m^2$  and  $2mn$ .

**Example.** We would like to find the rational point. For example,  $ax^2 + by^2 - c = 0$  with  $a, b, c \in \mathbb{Q}$ . We could rescale the denominator to assume  $a, b, c \in \mathbb{Z}$  and consider  $aX^2 + bY^2 + cZ^2 = 0$ . We want to find  $(X, Y, Z) \in \mathbb{Z}^3$  such that  $f(X, Y, Z) = 0$ . We have two observations:

- (1) If no solutions to  $F(X, Y, Z) = 0$  over  $\mathbb{R}$  then no solution over  $\mathbb{Z}$ .
- (2) If no solutions to  $F(X, Y, Z) \equiv 0 \pmod p$  with  $p$  prime, then no solutions over  $\mathbb{Z}$ .

**Example.**  $x^2 + y^2 = 3$ . We look at  $X^2 + Y^2 - 3Z^2 = 0$ . If the solution exists in  $\mathbb{Z}$ , then we must have  $X^2 + Y^2 \equiv 0 \pmod 3$ . But the only solution is the trivial solution  $(X, Y) \equiv (0, 0) \pmod 3$ . Given  $(X, Y, Z)$  solution, we know that  $X \equiv Y \equiv 0 \pmod 3$  then  $Z \equiv 0 \pmod 3$ . Then  $(\frac{X}{3}, \frac{Y}{3}, \frac{Z}{3})$  is a smaller solution. And we can repeat the process which implies  $(X, Y, Z) = (0, 0, 0)$ .

**Theorem 0.5 (Hasse–Minkowski theorem).** Let  $a, b, c \in \mathbb{Z}^\times$  and let  $F(X, Y, Z) = aX^2 + bY^2 + cZ^2$ . Then  $F(X, Y, Z)$  has rational solutions if and only if  $F$  has real solutions and  $F(X, Y, Z) \equiv 0 \pmod p$  has non-trivial mod  $p$  solutions.

Algorithm for finding rational points on a conic

- (1) Given a conic:

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

We transform it into  $ax^2 + by^2 + c = 0$  with all coefficients rational.

- (2) Apply Hasse–Minkowski theorem. If no solutions, we stop.
- (3) Otherwise, find an initial solution.
- (4) Use the "secant" method to produce all other rational points.

For cubics, the idea is that given 2 rational points on a rational cubic, form a rational line connecting the points. This line should intersect the cubic a third time producing a new rational points.

**Example.** Consider  $y^2 = x^3 - 25x$  which is equivalent to congruent number problem  $n = 5$ . Obvious rational points are  $(-5, 0), (0, 0), (5, 0)$  but these 3 points doesn't lead to new ones using the secant line construction. Let's map the triangle of congruent number problem  $n = 5$  to a point on the cubic to get a fourth rational point

$$(a, b, c) \mapsto (\frac{nb}{c-a}, \frac{2n^2}{c-a})$$

$$(\frac{3}{2}, \frac{20}{3}, \frac{41}{6}) \mapsto (\frac{25}{4}, \frac{75}{8}) = P$$

We can combine  $(-5, 0), (0, 0), (5, 0)$  with  $P$  to get new points. The line through  $P$  and  $(5, 0)$  is  $y = \frac{15}{2}(x - 5)$ . Intersect  $y^2 = x^3 - 25x$  we get

$$\frac{225}{4}(x - 5)^2 = x^3 - 25x \implies x^3 - \frac{225}{4}x^2 + \dots = 0$$

Clearly, the other root is 45. Thus, the other intersection is  $(45, 300)$ . Now, using

$$(x, y) \mapsto (\frac{x^2 - n^2}{y}, \frac{2nx}{y}, \frac{x^2 + n^2}{y})$$

$$(45, 300) \mapsto (\frac{20}{3}, \frac{3}{2}, \frac{41}{6})$$

New points, but same triangle.

Alternatively, what about combining  $P$  with itself by looking at the tangent. Expect that this intersects  $y^2 = x^3 - 25x$  a third time. The result is

$$(\frac{1681}{144}, \frac{62279}{1728})$$

which corresponds to the triangle

$$\left(\frac{1519}{492}, \frac{4920}{1519}, \frac{3344161}{747348}\right)$$

which has area 5.

For conics, the existence of an initial rational point follows from Hasse-Minkowski theorem. There is no corresponding result for cubic.

**Example (Selmer).** The curve  $3X^3 + 4Y^3 + 5Z^3 = 0$  has

- (1) Solutions over  $\mathbb{R}$ .
- (2) Solutions over mod  $m$  for any  $m \geq 1$ .

Yet,  $3X^3 + 4Y^3 + 5Z^3 = 0$  has only the trivial solution over  $\mathbb{Q}$ .

**Exercise.**

- (1) Find all rational points on the hyperbola  $x^2 - 5y^2 = 4$ .
- (2) Consider  $5X^2 - 2Y^2 - 5Z^2 = 0$ . Find at least one non-trivial rational point on this point.
- (3) Let  $a, b, c \in \mathbb{Z} \setminus \{0\}$ . Prove that  $ax^2 + by^2 = c$  has solution mod  $p$  where  $p$  is prime, provided  $p \nmid abc$ .

# 1 Elliptic Curves

**Definition 1.1.** An elliptic curve is a non-singular plane curve (i.e. in 2 variables) of degree 3 containing at least 1 rational point.

**Definition 1.2.** A curve is non-singular at  $P$  if the curve has a well-defined tangent at  $P$ . If the curve is defined by  $f(x, y) = 0, f(x, y) \in \mathbb{Q}[x, y]$ . Then tangent line at  $P = (x_P, y_P)$  is

$$\frac{\partial f}{\partial x}|_P(x - x_P) + \frac{\partial f}{\partial y}|_P(y - y_P) = 0$$

A curve is non-singular if and only if  $\nabla f(x, y) = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$  is non-zero for all  $(x, y)$  on the curve.

**Example.**  $x^3 + y^3 = 1$  contains  $(1, 0)$ . Let  $f(x, y) = x^2 + y^2 - 1$  then  $\nabla f(x, y) = (3x^2, 3y^2)$  which is zero if and only if  $x = y = 0$  which does not lie on  $x^3 + y^3 = 1$  so no problem here. Thus,  $x^3 + y^3 = 1$  is an elliptic curve.

**Examples (Non-examples).**

- (1)  $y^2 = x^3$ . Let  $f(x, y) = y^2 - x^3$  then  $\nabla f(x, y) = (-3x^2, 2y)$  which is zero at  $(0, 0)$  but this point lies on the curve. Thus, this is not an elliptic curve.
- (2)  $y^2 = x^2(x + 1)$ . Let  $f(x, y) = y^2 - x^3 - x^2$  then  $\nabla f(x, y) = (-3x^2 - 2x, 2y)$  which is singular at  $(0, 0)$ .

**Remark.** Finding  $\mathbb{Q}$  points on singular cubic curves is similar to the process for conics.

**Example.**  $y^2 = (x + 2)(x - 1)^2$  which has singular point at  $(1, 0)$ . Let  $z = \frac{y}{x-1}$  then  $z^2 = x + 1$  which is a parabola. The  $\mathbb{Q}$ -points look like  $(x, z) = (t^2 - 2, t) : t \in \mathbb{Q}$ . Thus,  $y = z(x - 1) = t(t^2 - 3)$  i.e.

$$(x, y) = (t^2 - 2, t(t^2 - 3)) : t \in \mathbb{Q}$$

**Theorem 1.3 (Bézout's Theorem, 1779, False Version).** Two plane curves of degree  $m$  and  $n$  should intersect in  $mn$  points.

**Example (Counterexample for the theorem above).**

- (1) Consider  $y = x, y = x$  then two curves intersect at infinite many times.
- (2) Consider  $y = x, y^2 = x^2$  then again we have infinite many intersections.

**Definition 1.4.** An irreducible curve is a curve  $f(x, y) = 0$  with  $f$  irreducible over  $\mathbb{C}[x, y]$ .

**Definition 1.5.** If  $f(x, y) = 0$  with  $f$  factorising  $f_1(x, y)f_2(x, y) = 0$ ,  $f_1, f_2$  irreducible,  $f_1$  and  $f_2$  are irreducible components of  $f$ .

Curve in Bézout's theorem should have no irreducible component in common.

**Example.**  $y = 0, y = x^2$  joint 1 intersection but it's a "double intersection".

When 1 curve is a line,  $y = ax + b$  and the other is 0. Intersection multiplicity can be defined as the multiplicity of root of  $f(x, ax + b) = 0$  at the intersection.

**Example.**  $y = 0$  and  $y = x^2 + 1$ . Intersections at  $(i, 0), (-i, 0)$  but Bézout's Theorem only holds for algebraically closed fields.

**Example.** Consider  $y = x, y = x + 1$

**Definition 1.6** (Projective Geometry). Let  $k$  be a field and  $k^{n+1} = \{(x_0, \dots, x_n) : x_i \in k\}$ . For integer  $n \geq 1$ , define projective  $n$ -space by

$$\mathbb{P}^n(k) = \{k^{n+1} \setminus \{0, \dots, 0\}\} / \sim$$

where the equivalence relation  $\sim$  is given by

$$(x_0, \dots, x_n) \sim (y_0, \dots, y_n) \text{ iff } (y_0, \dots, y_n) = (\lambda x_0, \dots, \lambda x_n)$$

for some  $\lambda \neq 0$  in  $k$ .

The coset containing  $(x_0, \dots, x_n)$  is usually notated in capital term

$$[X_0 : X_1 : \dots : X_n]$$

called projective coordinate. On contrast,  $(x_0, \dots, x_n)$  are called affine coordinates. Affine  $n$ -space is  $k^n$ , is sometimes written  $\mathbb{A}^n(k)$ .

To do computation in  $\mathbb{P}^k$  use affine curve: Consider  $\mathbb{P}^2(\mathbb{Q})$  with coordinate  $[X : Y : Z]$ . Suppose  $Z \neq 0$ , let  $x = \frac{X}{Z}$  and  $y = \frac{Y}{Z}$ . We know that

$$[X : Y : Z] = [x : y : 1]$$

In a region away from  $Z = 0$ , we have a bijection from projective coordinates  $[X : Y : Z]$  to affine coordinates  $(x, y) \in \mathbb{A}^2$ . Likewise, if  $X \neq 0$  let  $y = \frac{Y}{X}$  and  $z = \frac{Z}{X}$  then

$$[X : Y : Z] \sim [1 : y : z] \cong (y, z) \in \mathbb{A}^2$$

Given any  $[X : Y : Z]$ , there exists a non-zero coordinate. Scale by that coordinate, and do computation in affine space.

**Definition 1.7.** Given  $F(X, Y, Z) \in \mathbb{Q}[X, Y, Z]$  homogeneous,  $F(X, Y, Z) = 0$  defines a subset of  $\mathbb{P}^2(\mathbb{Q})$  called a projective curve.

**Definition 1.8.** Given a curve  $f(x, y) = 0$ , we define the projective completion of  $f$  as

$$Z^d F\left(\frac{X}{Z}, \frac{Y}{Z}\right) = 0 \text{ where } d = \deg f$$

**Example.**  $y = x \rightarrow f(x, y) = y - x$  and  $y = x + 1 \rightarrow g(x, y) = y - x - 1$ . Then

$$F(X, Y, Z) = Z\left(\frac{Y}{Z} - \frac{X}{Z}\right) = Y - X$$

$$G(X, Y, Z) = Z\left(\frac{Y}{Z} - \frac{X}{Z} - 1\right) = Y - X - Z$$

$F(X, Y, Z) = G(X, Y, Z) = 0$  if  $Y = X$  and  $Y = X + Z$ . Thus,  $Z = 0$  and  $Y = X$ .

$$[X : Y : Z] = [X : X : 0] = [1 : 1 : 0]$$

which is the unique intersection. These kinds of points are called "points at infinity".

**Example.** Consider  $y^2 = x^2 + 1$  and  $y = x$ . Then

$$F(X, Y, Z) = Y^2 - X^2 - Z^2$$

$$G(X, Y, Z) = Y - X$$

At intersections,  $Y = X, Z = 0$ . We get an intersection at  $[X : Y : Z] = [1, 1, 0]$ . Use  $[X : Y : Z] = [\frac{X}{Y} : 1 : \frac{Z}{Y}] = [x : 1 : z]$ . Then

$$F(X, Y, Z) = Y^2 - X^2 - Z^2 = Y^2(1 - x^2 - z^2)$$

$$G(X, Y, Z) = Y - X = Y(1 - x)$$

$1 - x^2 - z^2 = 1 - x = 0$  and  $x = 1$  is the tangent line of  $1 - x^2 - z^2$  so multiplicity is 2.

**Theorem 1.9** (Bézout's Theorem). Let  $C_1$  and  $C_2$  be two projective plane curves with no irreducible components in common. Then  $C_1$  and  $C_2$  intersect in degree  $\deg C_1 \deg C_2$  points over  $\mathbb{C}$ , counted with multiplicity.

Let  $P_1, P_2$  be rational points on  $E$  denoted  $E(\mathbb{Q})$ . Bézout's theorem implies  $P_1$  and  $P_2$  intersects  $E$  a third time, call this point  $P_1 * P_2$ .  $*$  :  $E(\mathbb{Q}) \times E(\mathbb{Q}) \rightarrow E(\mathbb{Q})$  is a binary operation which is not a group operation since there's no identity. To make a group, first fix a base point. Fix a point  $O \in E(\mathbb{Q})$  which will become identity. The third intersection of the line passing through  $P_1 * P_2$  and  $O$  with the curve is defined by  $P_1 + P_2, (P_1 + P_2 = O * (P_1 * P_2))$ .

**Theorem 1.10.** The set of rational points on an elliptic curve form an abelian group.

**Proof.** We show each criteria of an abelian group:

(i) Commutativity:  $P_1 + P_2 = O * (P_1 * P_2) = O * (P_2 * P_1) = P_2 * P_1$ .

(ii) Identity: Let  $O$  be the identity. We want to show  $O + P = P$  for all  $P \in E(\mathbb{Q})$ . Let  $Q = O * P$  then

$$O + P = O * (O * P) = O * Q$$

which is the third point on line containing  $O$  and  $Q$  but we know that  $Q$  is the third point on line containing  $O$  and  $P$ . Thus,  $O + P = P$ .

(iii) Inverse: We want to show that given  $P$  there exists  $(-P)$  such that  $P + (-P) = O$ . First, define  $S$  as the third point of intersection between  $E$  and the tangent to  $E$  at  $O$ . Then  $S = O * O$  and we define  $-P = S * P$ . Verify

$$P + (-P) = O * (P * (-P)) = O * S = O$$

since at  $O$  the line is a tangent.

(iv) Associativity:  $P + (Q + R) = (P + Q) + R$  for all  $P, Q, R \in E(\mathbb{Q})$ . We want to show  $(P + Q) * R = P * (Q + R) = U$ . Let  $C_1$  be the reducible cubic given by union of horizontal lines.  $C_2$  be the reducible cubic given by the union of vertical lines.  $C_1$  and  $C_2$  intersect in

$$P, U, Q + R, Q, R, Q * R, P * Q, P + Q, O$$

Also,  $E$  intersects all of these except maybe  $U$ . By the Lemma below,  $E$  contains  $U$ . The line containing  $P + Q, R, U, (P + Q) * R$  intersects  $E$  4 times. By Bézout's,  $U = (P + Q) * R$ . Similarly, we can show that  $U = P * (Q + R)$ . Since  $(P + Q) * R = P * (Q + R)$ .  $O * ((P + Q) * R) = O * (P * (Q + R))$  i.e.

$$(P + Q) + R = P + (Q + R)$$

□

**Lemma 1.11.** Let  $C_1, C_2$  be two cubic curves intersecting in  $P_1, \dots, P_9$  counted with multiplicity. Any cubic curve passing through 8 of these points also passes through the 9<sup>th</sup>.

**Proof.** Let  $P_i = [X_i : Y_i : Z_i]$ ,  $i \leq 9$ . Assume that the vectors

$$(X_i^3, X_i^2 Y_i, X_i^2 Z_i, X_i Y_i^2, X_i Y_i Z_i, X_i Z_i^2, Y_i^3, Y_i^2 Z_i, Y_i Z_i^2, Z_i^3) \in \mathbb{C}^{10}$$

are linearly independent. Note: any cubic has the form  $a_1 X^3 + a_2 X^2 Y + \dots + a_{10} Z^3 = 0$  is a 10-dimensional space. Every time I impose a condition like  $P_1$  on the curve, I get a linear condition on the  $a_i$ 's. The vector space  $V$  of vector equation of curve equations for curve intersecting  $P_1, \dots, P_8$  is  $10 - 8 = 2$ -dimensional.  $V$  contains equations  $F_1 = 0$  for  $C_1$  and  $F_2 = 0$  for  $C_2$ . In general, my cubic curve passing through  $P_1, \dots, P_8$  is defined by  $F = \lambda_1 F_1 + \lambda_2 F_2 = 0$ .  $F(P_9) = \lambda_1 F_1(P_9) + \lambda_2 F_2(P_9) = \lambda_1 0 + \lambda_2 0 = 0$ . Thus,  $P_9$  lies on the curve determined by  $F = 0$ . □

## 2 Weierstrass Form

Cubic plane curve:

$$a_1 x^3 + a_2 x^2 + a_3 x + a_4 + a_5 x^2 y + a_6 x y + a_7 y + a_8 x y^2 + a_9 y^2 + a_{10} y^3 = 0$$

10-coordinates which makes computation unwieldy. We want: Some analogue of "canonical form" for conics,  $ax^2 + by^2 = c$ , which reduces the number of monomials.

**Definition 2.1.** A cubic curve of the form

$$y^2 = x^3 + ax + b, \quad a, b \in \mathbb{Q}$$

is said to be in Weierstrass normal form.

**Definition 2.2.** Two curves are birationally equivalent if there exists a map  $f : C_1 \rightarrow C_2$  where  $f$  is expressed by rational functions in each coordinate. And the same is true for  $f^{-1} : C_2 \rightarrow C_1$ .

**Example.** The circle  $x^2 + y^2 = 1$  is birationally equivalent to the projective line  $\mathbb{P}^1(\mathbb{R}) = \{[a : b]\}$ . Since points on the circle are parametrised by

$$\left( \frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2} \right), \quad t \in \mathbb{R}$$

misses  $(-1, 0)$  and this corresponds to  $[1 : 0]$ .

**Theorem 2.3.** Any elliptic curve over  $\mathbb{Q}$  is birationally equivalent to an elliptic curve in Weierstrass form.

One of the main steps in turning a curve into Weierstrass form is a transformation called a projective linear map.

**Definition 2.4.** A projective linear map is a map  $T_A : [X : Y : Z] \rightarrow [X' : Y' : Z']$  where  $\begin{pmatrix} X' \\ Y' \\ Z' \end{pmatrix} = A \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$  for some matrix  $A \in GL_3(\mathbb{Q})$ .

**Remark.**

- (1)  $T_A$  is well-defined.
- (2)  $T_A = T_B$  if and only if  $A = \lambda B, \lambda \in \mathbb{Q}$ .
- (3)  $T_A$  maps projective lines to projective lines.
- (4) If  $C : F(X, Y, Z) = 0$ , the image of  $C$  under  $T_A$ , call it  $T_A(C)$  is  $T_A(C) : F(X', Y', Z') = 0$ .
- (5)  $T_A$  preserves degree.

**Example.**  $X^3 + Y^3 = 2Z^3$ . Apply a projective linear map, map 3 significant lines to the coordinate axes.

- (a)  $L_1$  is the tangent line to the curve at  $O$ . Take  $O = [1 : 1 : 1]$  and  $P = O * O$ .
- (b)  $L_2$  is the tangent line to curve at  $P$ .
- (c)  $L_3 \neq L_1$  passing through  $O$ .

Let  $F(X, Y, Z) = X^3 + Y^3 - 2Z^3$  then  $\nabla F(X, Y, Z) = (3X^2, 3Y^2, -6Z^2)$ ,  $\nabla F(1, 1, 1) = (3, 3, -6)$  i.e.  $L_1 : X + Y - 2Z = 0$ . We intersect  $X + Y - 2Z = 0$  and  $X^3 + Y^3 = 2Z^3$ , if  $Z = 0$  then  $X + Y = 0$  and  $X^3 + Y^3 = 0$  gives solution  $[X : Y : Z] = [X : -X : 0] = [1 : -1 : 0]$ . Thus,  $P = [1, -1, 0]$ . Map  $L_1$  to  $Z_1 = 0$  ( $Z_1 = X + Y - 2Z$ ).

$\nabla F(P) = (3, 3, 0)$  so  $L_2 : 3X + 3Y + 0Z = 0 \implies X + Y = 0$ . Map to  $X_1 = 0$  ( $X_1 = X + Y$ ). For  $L_3$ , choose  $X - Y = 0$  for simplicity.  $Y_1 = X - Y$ . In matrix form,

$$\begin{pmatrix} X_1 \\ Y_1 \\ Z_1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 1 & 1 & -2 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} \\ \implies \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} X_1 \\ Y_1 \\ Z_1 \end{pmatrix}$$

Substitute into the original curve  $X^3 + Y^3 = 2Z^3$  we get

$$\begin{aligned} \left(\frac{1}{2}X_1 + \frac{1}{2}Y_1\right)^3 + \left(\frac{1}{2}X_1 - \frac{1}{2}Y_1\right)^3 &= 2\left(\frac{1}{2}X_1 - \frac{1}{2}Z_1\right)^3 \\ X_1^3 + 3X_1^2Y_1 + 3X_1Y_1^2 + Y_1^3 + X_1^3 - 3X_1^2Y_1 + 3X_1Y_1^2 - Y_1^3 &= 2X_1^3 - 6X_1^2Z_1 + 6X_1Z_1^2 - 2Z_1^3 \\ 6X_1Y_1^2 &= -6X_1^2Z_1 + 6X_1Z_1^2 - 2Z_1^3 \\ \implies 3X_1Y_1^2 + 3X_1^2Z_1 - 3X_1Z_1^2 + Z_1^3 &= 0 \end{aligned}$$

Done with projective linear map. Convert to affine  $(x_1, y_1)$  we have

$$3x_1y_1^2 + 3x_1^2 - 3x_1 + 1 = 0$$

$y_1$  only appears in  $x_1y_1^2$  we have

$$x_1^2y_1^2 = -x_1^3 + x_1^2 - \frac{1}{3}x_1$$

Let  $y_2 = x_1y_1$  then

$$y_2^2 = -x_1^3 + x_1^2 - \frac{1}{3}x_1$$

Let  $x_2 = -x_1$  to fix leading coefficient  $y_2^2 = x_2^3 + x_2^2 + \frac{1}{3}x_2$ . Shift to make  $x_2^2$  coefficient equals to zero.  $x_3 = x_2 + d$  for some  $d$ ,

$$\begin{aligned} y_2^2 &= (x_3 - d)^3 + (x_3 - d)^2 + \frac{1}{3}(x_3 - d) \\ &= x_3^3 - 3x_3^2d + 3x_3d^2 - d^3 + x_3^2 - \dots \end{aligned}$$

We want  $(-3d + 1)x_3^2 = 0$  i.e.  $d = \frac{1}{3}$  then we have

$$y_2^2 = x_3^3 - \frac{1}{27}$$

which is the Weiestrass form.

**Remark (Applications of Weiestrass Form).**

- (1)  $y^2 = x^3 + ax + b$ , let  $f(x, y) = y^2 - x^3 - ax - b$  then  $\nabla f(x, y) = (3x^2 + a, -2y)$ . If  $\nabla f(x, y) = 0$  we have

$$3x^2 + a = 0, \text{ and } y = 0$$

If  $x^3 + ax + b$  has double root then  $3x^2 + a = 0$ . Thus,  $y^2 = x^3 + ax + b$  is non-singular if and only if  $x^3 + ax + b = 0$  has distinct root. Let  $g(x) \in \mathbb{C}[x]$ ,  $\lambda_1, \dots, \lambda_d$  be the roots of  $g$ . Let

$$\Delta(g) = \prod_{i \neq j} (\lambda_i - \lambda_j) = \prod_{i < j} (\lambda_i - \lambda_j)^2$$

$\Delta(g)$  is the discriminant of  $g$  note that  $\Delta(g) = 0$  if and only if there is repeated root. In fact, if  $g(x) = x^3 + ax + b$  we have  $\Delta(g) = -4a^3 - 27b^2$ .

- (2) It's easy to use affine coordinates for curves in Weiestrass form.  $y^2 = x^3 + ax + b$ , projective completion:

$$Y^2Z = X^3 + aXZ^2 + bZ^3$$

The points we "miss" in the  $(x, y)$  affine coordinates are where  $Z = 0$  i.e.  $X = 0$ . Thus, we miss just one point  $[0 : 1 : 0]$  which is the point  $O$ .

- (3) Simplifications to group law: If the curve is in Weiestrass form: Then  $P + Q$  is the intersection of vertical line passing through  $P * Q$  and the curve. For a curve in Weiestrass form,  $P + Q$  is equal to the reflection of  $P * Q$  over the  $x$ -axis.
- (4) Points of order 2, called "2-torsion" i.e.  $P + P = O$ .  $P = -P$ . Reflection of  $P$  over  $x$ -axis. Thus,  $P$  has order 2 if and only if  $P = (x, y)$  with  $y = 0$ . For  $y^2 = x^3 + ax + b$  we could have 0, 1, 3 points of order 2.

## 3 Elliptic Functions

**Definition 3.1.** Fix  $\omega_1, \omega_2 \in \mathbb{C}$  linearly independent over  $\mathbb{R}$ . The set

$$\Lambda = \{m_1\omega_1 + m_2\omega_2 : m_1, m_2 \in \mathbb{Z}\}$$

is called a lattice, denoted by  $\Lambda(\omega_1, \omega_2)$ .

**Remark.**

- (1)  $\{\omega_1, \omega_2\}$  is called a basis for  $\Lambda$ .
- (2)  $\Lambda(\omega_1, \omega_2)$  may equal  $\Lambda(\omega'_1, \omega'_2)$  i.e. basis isn't unique. In fact, they are related by action of  $SL_2(\mathbb{Z})$ .

**Definition 3.2.** Given  $\Lambda(\omega_1, \omega_2)$ , the region

$$\{x_1\omega_1 + x_2\omega_2, \quad 0 \leq x_1, x_2 \leq 1\}$$

is the fundamental parallelogram.

**Definition 3.3.** An elliptic function with respect to  $\lambda$  is a meromorphic function  $f : \mathbb{C} \rightarrow \hat{\mathbb{C}}$  satisfying

$$f(z + \omega) = f(z), \quad z \in \mathbb{C}, \omega \in \Lambda$$

$f$  is elliptic if and only if  $f(z + \omega_1) = f(z)$ ,  $f(z + \omega_2) = f(z)$ . For this reason, elliptic functions are sometimes called "doubly-periodic".

Goal: Find (non-constant) elliptic function,

**Theorem 3.4 (Liouville).** Let  $f$  be elliptic with respect to  $\Lambda$ .

- (1) If  $f$  is holomorphic,  $f$  is constant.
- (2) Let  $P(f) =$  set of poles of  $f$  in  $\mathbb{C}$ .

$$\sum_{z_0 \in P(f) \cap P} \text{Res}_{z=z_0} f(z) = 0,$$

where  $P$  is the fundamental parallelogram.

- (3) Let  $\text{ord}(z; f) =$  order of pole/zero of  $f$ . Then

$$\sum_{z_0 \in P(f) \cap P} \text{ord}(z, f) = \sum_{z_0 \in Z(f) \cap P} \text{ord}(z_0; f)$$

i.e. same number of zeros and poles.

**Proof.**

- (1) Let  $f$  be holomorphic, on  $P$ ,  $f$  is bounded. By periodicity,  $f$  is bounded on  $\mathbb{C}$ . Thus,  $f$  is constant.
- (2) Want to use Cauchy's Residue Theorem. Suppose all poles of  $P$  lie in the interior of  $P$ .

$$\begin{aligned} 2\pi i \sum_{z_0 \in P(f) \cap P} \text{Res}_{z=z_0} f(z) &= \oint_{\partial P} f(z) dz \\ &= \int_{C_1} f(z) dz + \int_{C_2} f(z) dz + \int_{C_3} f(z) dz + \int_{C_4} f(z) dz \end{aligned}$$

Hence,

$$\int_{C_1} = -\int_{C_3}, \quad \int_{C_2} = -\int_{C_4}$$

by periodicity of  $f$ .

- (3) Want an auxiliary  $f$  whose poles have residues giving the order of poles/zeros of  $f$ . Use  $g(z) := \frac{f'(z)}{f(z)}$ , logarithm

derivative.  $g$  has poles at poles/zeros of  $f$ .

$$\operatorname{Res}_{z=z_0, z_0 \in P(g)} g(z) = \begin{cases} \text{order of pole of } f \\ -\text{order of zero of } f \end{cases}$$

Using Cauchy,

$$2\pi i \sum_{z_0 \in P \cap P(g)} \operatorname{Res}_{z=z_0} g(z) = \int_{\partial P} g(z) dz$$

RHS is zero since  $g(z)$  is elliptic

$$LHS = 2\pi i \left( \sum_{z_0 \in P(f) \cap P} \operatorname{ord}(z_0; P) - \sum_{z_0 \in Z(f) \cap P} \operatorname{ord}(z_0; f) \right)$$

□

**Corollary 3.5.** Simple non-constant elliptic functions have either

- (i) 1 double pole.
- (ii) 2 simple poles, opposite residue.

**Remark.** (1),(2),(3) are necessary conditions, but not sufficient. But Abel gives the sufficient condition.

**Theorem 3.6.** Let  $\{Z_1, \dots, Z_k\}, \{P_1, \dots, P_k\}$  be disjoint multi-sets contained in  $P$ . There exists an elliptic function for  $\Lambda$  with zeros at  $Z_j$  (with multiplicity) and poles at  $P_j$  (with multiplicity) if and only if

$$\sum_j Z_j - \sum_j P_j \in \Lambda$$

Warmup: Constructing 1-periodic function  $f : \mathbb{R} \rightarrow \mathbb{R}$ . Starting with  $g : \mathbb{R} \rightarrow \mathbb{R}$ . Consider  $f(x) := \sum_{n \in \mathbb{Z}} g(x+n)$ . RHS converges if  $g$  has sufficient decay near  $\infty$ ,

$$g(x) = O(|x|^{-1-\varepsilon}), \varepsilon > 0$$

$f(x+1) = \sum_{n \in \mathbb{Z}} g(x+1+n) = \sum_{n' \in \mathbb{Z}} g(x+n') = f(x)$ . Get periodicity through averaging.

Warmup 2: Construct a meromorphic  $f : \mathbb{C} \rightarrow \hat{\mathbb{C}}$ , 1-periodic with poles only at  $n \in \mathbb{Z}$ . Take  $g(z) = \frac{1}{z}$  then

$$\sum_{n \in \mathbb{Z}} \frac{1}{z+n} \text{ but not absolutely convergent}$$

Fix

$$f(z) = \frac{1}{z} + \sum_{\substack{n \in \mathbb{Z} \\ n \neq 0}} \left( \frac{1}{z+n} - \frac{1}{n} \right)$$

Since

$$\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$$

$f(z)$  converges for fixed  $z \in \mathbb{C} \setminus \{\mathbb{Z}\}$ .  $f(z)$  converges to a holomorphic function on  $\mathbb{C} \setminus \mathbb{Z}$ . Consider  $f'(z)$ ,

$$f'(z) = -\frac{1}{z^2} + \sum_{\substack{n \in \mathbb{Z} \\ n \neq 0}} -\frac{1}{(z+n)^2} = -\sum_{n \in \mathbb{Z}} \frac{1}{(z+n)^2}$$

implies that  $f'(z+1) - f'(z) = 0$  i.e  $f(z+1) - f(z) = C$ . Indeed,  $f(z)$  is periodic if  $C = 0$ . Consider

$$f\left(\frac{1}{2}\right) = 2 + \sum_{\substack{n \in \mathbb{Z} \\ n \neq 0}} \left( \frac{1}{n + \frac{1}{2}} - \frac{1}{n} \right)$$

$$f\left(-\frac{1}{2}\right) = -2 + \sum_{\substack{n \in \mathbb{Z} \\ n \neq 0}} \left( \frac{1}{n - \frac{1}{2}} - \frac{1}{n} \right)$$

Thus,

$$\begin{aligned} f\left(\frac{1}{2}\right) - f\left(-\frac{1}{2}\right) &= 4 + \sum_{\substack{n \in \mathbb{Z} \\ n \neq 0}} \left( \frac{1}{n + \frac{1}{2}} - \frac{1}{n} \right) - \sum_{\substack{n \in \mathbb{Z} \\ n \neq 0}} \left( \frac{1}{n - \frac{1}{2}} - \frac{1}{n} \right) \\ &= 4 + \sum_{n \geq 0} \left( \frac{1}{n + \frac{1}{2}} - \frac{1}{n - \frac{1}{2}} \right) \\ &= \sum_{n \in \mathbb{Z}} \left( \frac{1}{n + \frac{1}{2}} - \frac{1}{n - \frac{1}{2}} \right) = 0 \end{aligned}$$

To show it's meromorphic, by periodicity. It suffices to show it's meromorphic at  $z = 0$ ,

$$f(z) = \frac{1}{z} + \sum_{n \neq 0} \left( \frac{1}{z+n} - \frac{1}{n} \right)$$

Near  $z = 0$ , only  $\frac{1}{z}$  is non-holomorphic so  $f(z)$  is meromorphic since  $\frac{1}{z}$  is.

**Remark.**  $f(z) = \pi \cot(\pi z)$ .

Constructing Elliptic Functions. Let  $\Lambda = \Lambda(\omega_1, \omega_2)$ . Idea: Get double periodicity by averaging over  $\Lambda$ .

$$\sum_{\omega \in \Lambda} g(z + \omega) \text{ for some } g$$

By Liouville, consider  $g$  with a double pole  $g(z) = \frac{1}{z^2}$ . Construct

$$\sum_{\omega \in \Lambda} \frac{1}{(z + \omega)^2}$$

But this doesn't converge. Modify:

$$\frac{1}{z^2} + \sum_{\substack{\omega \in \Lambda \\ \omega \neq 0}} \left( \frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right)$$

this is called Weierstrass  $\wp$ -function. Depends on  $\Lambda$ , but usually suppose this dependence.

Claim:  $\wp$  is elliptic on  $\Lambda$ .

(1) Converge absolutely.

$$\begin{aligned} \frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} &= \frac{\omega^2}{\omega^2(z + \omega)^2} - \frac{(z + \omega)^2}{\omega^2(z + \omega)^2} \\ &= O\left(\frac{1}{|\omega|^2}\right) \end{aligned}$$

(2) To see the double pole at  $z = 0$ ,

$$\wp(z) - \frac{1}{z^2} = \sum_{\omega \neq 0} \left( \frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right)$$

$\implies z = 0$  is a double pole since RHS is holomorphic near  $z = 0$ . Similar to other poles.

(3) Periodicity:

$$\wp'(z) = -\frac{2}{z^3} + \sum_{\substack{\omega \neq 0 \\ \omega \in \Lambda}} -\frac{2}{(z + \omega)^3}$$

Check that the series converges absolutely:

$$\wp'(z) = -2 \sum_{\omega \in \Lambda} \frac{1}{(z + \omega)^3}$$

Clearly periodic:  $\wp'(z + \omega) = \wp'(z)$ ,  $\omega \in \Lambda$ . Thus,  $\wp(z + \omega) - \wp(z) = C$ . Evaluate at  $z = -\frac{\omega}{2}$  where  $\omega$  is one of the basis elements of  $\Lambda$ .

$$\wp\left(\frac{\omega}{2}\right) - \wp\left(-\frac{\omega}{2}\right) = C$$

But note that  $\wp(z)$  is even, thus  $C = 0$ .

Laurent expansion of  $\wp(z)$  at  $z = 0$ . We know  $\wp(z)$  has a pole  $\frac{1}{z^2}$ , want further terms.

**Proposition 3.7.**

$$\wp(z) = \frac{1}{z^2} + \sum_{k=1}^{\infty} (2k+1) G_{2k+2}(\Lambda) z^{2k}$$

where

$$G_{2k}(\Lambda) := \sum_{\substack{\omega \in \Lambda \\ \omega \neq 0}} \omega^{-2k}$$

and  $G_{2k}(\Lambda)$  is called the Eisenstein series of weight  $2k$ .

**Proof.** By definition,

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \neq 0} \left( \frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right)$$

Consider

$$\frac{1}{1-u} = \sum_{r=0}^{\infty} u^r \implies \frac{1}{(1-u)^2} = \sum_{r=1}^{\infty} r u^{r-1}$$

$$\begin{aligned}
\frac{1}{(z+\omega)^2} &= \frac{1}{\omega^2} \cdot \frac{1}{(z/\omega)^2} = \frac{1}{\omega^2} \sum_{r=1}^{\infty} r \left(-\frac{z}{\omega}\right)^{r-1} \\
&= \sum_{r=1}^{\infty} \frac{r(-1)^{r-1} z^{r-1}}{\omega^{r-1}} \\
\Rightarrow \frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} &= \sum_{r=2}^{\infty} \frac{r(-1)^{r-1} z^{r-1}}{\omega^{r-1}}
\end{aligned}$$

Now sum over  $\omega \in \Lambda$ ,  $\omega \neq 0$ :

$$\sum_{\omega \neq 0} \left( \frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} \right) = \sum_{r=2}^{\infty} r(-1)^{r-1} z^{r-1} \sum_{\omega \neq 0} \omega^{-r-1}$$

Let  $k = r - 1$ ,

$$\sum_{k=1}^{\infty} (k+1)(-1)^k \left( \sum_{\omega \neq 0} \omega^{-k-2} \right) z^k.$$

Note that for  $k$  odd,  $\sum_{\omega \neq 0} \omega^{-k} = 0$  since contributions from  $\omega$ ,  $-\omega$  cancel. Thus,

$$\wp(z) = \frac{1}{z^2} + \sum_{k=1}^{\infty} (2k+1) G_{2k+2}(\Lambda) z^{2k}$$

□

**Theorem 3.8.**  $\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3$  where  $g_2 = 60G_4(\Lambda)$ ,  $g_3 = 140G_6(\Lambda)$ .

**Proof.** Let  $F(z) = \wp'(z)^2 - 4\wp(z)^3 + g_2\wp(z) + g_3$ . We first show that  $F(z)$  is holomorphic. The only potential poles of  $F$  are at  $z \in \Lambda$ .  $F$  is elliptic, so it suffices to consider the pole at  $z = 0$ . Compute Laurent expansion of  $F$  at  $z = 0$ .

$$\begin{aligned}
\wp(z) &= \frac{1}{z^2} + 3G_4z^2 + 5G_6z^4 + 7G_8z^6 + O(z^8) \\
\wp(z)^2 &= z^{-4} + 6G_4 + 10G_6z^2 + O(z^4) \\
\wp(z)^3 &= z^{-6} + (6G_4 + 3G_4)z^{-2} + (5G_6 + 10G_6) + O(z^2) \\
\wp'(z) &= -2z^{-3} + 6G_4z + 20G_6z^3 + O(z^5) \\
\wp'(z)^2 &= 4z^{-6} - 24G_4z^{-2} - 80G_6 + O(z^2) \\
F(z) &= 4z^{-6} - 24G_4z^{-2} - 80G_6 + O(z^2) \\
&\quad - 4z^{-6} - 36G_4z^{-2} - 60G_6 + O(z^2) \\
&\quad + 60G_4z^{-2} + 180G_4^2z^2 + O(z^4) \\
&\quad + 140G_6 = 0 + O(z^2)
\end{aligned}$$

Thus, removable singularities at  $z = 0$ . Also,  $\lim_{z \rightarrow 0} F(z) = 0$ , by Laurent expansion. Therefore,  $F(0) = 0 \Rightarrow F(z) = 0$  for all  $z$ . □

Let

$$\begin{aligned}
\Phi : \mathbb{C}/\Lambda &\rightarrow E(\Lambda) \\
z \pmod{\Lambda} &\mapsto \begin{cases} (p(z), p'(z)), & z \notin \Lambda \\ 0 & z \in \Lambda \end{cases}
\end{aligned}$$

**Theorem 3.9 (Uniformization Theorem).** Let  $\Lambda$  be a lattice. Then

- (1)  $E(\Lambda)$  is an elliptic curve.
- (2)  $\Phi : \mathbb{C}/\Lambda \rightarrow E(\Lambda)$  is a group homomorphism.

We would show that  $E(\Lambda)$  is non-singular,  $\Phi$  is bijection and then  $\Phi$  is homomorphism.

(1)  $E(\Lambda)$  is non-singular if and only if  $4x^3 - g_2x - g_3$  has distinct roots. Let  $\Lambda = \Lambda(\omega_1, \omega_2)$ , define  $\omega_3 = \omega_1 + \omega_2$

**Proposition 3.10.**

- (1)  $\wp(\frac{\omega_1}{2}), \wp(\frac{\omega_2}{2}), \wp(\frac{\omega_3}{2})$  are all distinct.
- (2)  $\wp'(z)^2 = 4 \prod_{j=1}^3 (\wp(z) - \wp(\frac{\omega_j}{2}))$

**Proof.**

- (1) Consider  $f(z) = \wp(z) - \wp(\frac{\omega_1}{2})$ . Has a zero at  $\frac{\omega_1}{2}$ . If  $\wp(\frac{\omega_1}{2}) = \wp(\frac{\omega_2}{2})$ , then  $f$  has a zero at  $\frac{\omega_2}{2}$  as well. Since  $\frac{\omega_1}{2} \cdot 2 \in \Lambda$ , expected just a double zero at  $\frac{\omega_1}{2}$  by Lemma below. So zero at  $\frac{\omega_2}{2}$  is contradiction. Thus,  $\wp(\frac{\omega_1}{2}) \neq \wp(\frac{\omega_2}{2})$ . More generally,  $\wp(\frac{\omega_1}{2}), \wp(\frac{\omega_2}{2}), \wp(\frac{\omega_3}{2})$  are all distinct.

(2) Consider

$$F(z) = \frac{\wp'(z)^2}{4 \prod_{j=1}^3 (\wp(z) - \wp(\frac{\omega_j}{2}))}$$

We claim that  $F(z) \equiv 1$ . Let's show that  $F$  is holomorphic, potential poles: poles of numerator: pole at  $z = 0$ , order 6. Zeros of denominator: By Lemma below,  $\wp(z) - \wp(\frac{\omega_j}{2})$  has a zero of  $z \equiv \frac{\omega_j}{2} \pmod{\Lambda}$ , no other zeros. So we get 2 zeros at  $\frac{\omega_1}{2}, \frac{\omega_2}{2}, \frac{\omega_3}{2}$ . Each pole is cancelled in  $F(z)$ , at  $z = 0$

$$\frac{\text{order 6}}{(\text{order 2})(\text{order 2})(\text{order 2})} \implies \text{no poles}$$

At  $z = \frac{\omega_j}{2}$ , need to cancel order 2 pole in denominator. We claim that  $\wp'(\frac{\omega_j}{2}) = 0$ .  $\wp(z)$  is even, so  $\wp'(z)$  is odd.

$$\begin{aligned} \wp'(\frac{\omega_j}{2}) &= \wp'(\frac{\omega_j}{2} - \omega_j) = \wp'(-\frac{\omega_j}{2}) = -\wp'(\frac{\omega_j}{2}) \\ &\implies \wp'(\frac{\omega_j}{2}) = 0. \end{aligned}$$

Then  $F$  has no poles so  $F$  is constant say  $F = C$ . To find the constant, compare Laurent expansion at  $z = 0$ :

$$\begin{aligned} \wp(z) &= z^{-2} + 3G_4z^2 + 5G_6z^4 + \dots \\ \wp'(z) &= -2z^{-3} + 6G_4z \\ F(z) &= \frac{(-2z^{-3} + O(z))^2}{4(z^{-1} + O(1))^3} = \frac{4z^{-6} + O(z^{-2})}{4z^{-6} + O(z^{-4})} = \frac{4 + O(z^4)}{4 + O(z^2)} \end{aligned}$$

Let  $z \rightarrow 0$  we get  $F(z) = 1$ .

□

Motivation: Under Uniformization, points of order 2 maps to points of order 2.  $\frac{\omega_1}{2}, \frac{\omega_2}{2}, \frac{\omega_3}{2}$  are points of order 2 in  $\mathbb{C}/\Lambda$ . On the other hand, the points of order 2 on  $E(\Lambda)$  are  $(x, y)$  satisfies  $4x^3 - g_2x - g_3 = 0$  and  $y = 0$ .

**Lemma 3.11.** Let  $a \in \mathbb{C} \setminus \Lambda$ . Let  $f(z) = \wp(z) - \wp(a)$ . Then  $f$  is elliptic, and  $f$  has

- (i) double pole at  $z \equiv 0 \pmod{\Lambda}$ .
- (ii) zeros at  $\pm a \pmod{\Lambda}$ , if  $2a \notin \Lambda$  or a double zero at  $a$ , if  $2a \in \Lambda$ .

**Proof.** It's clear that  $f$  is elliptic and has a double pole at  $z \equiv 0 \pmod{\Lambda}$ . Obviously,  $f(a) = \wp(a) - \wp(a) = 0$  so  $a \pmod{\Lambda}$  is a zero. Moreover,  $f$  is even so  $f(-a) = 0$ . If  $a \not\equiv -a \pmod{\Lambda}$ , we have 2 distinct zeros. By Liouville, 2 poles implies 2 zeros so we found them all. If  $2a \in \Lambda$ ,  $a$  is a zero. Need a second zero  $b$ . Know that  $a + b - 2 \cdot 0 \in \Lambda$  by Abel's Theorem i.e.  $a + b \in \Lambda$ . Thus,  $a \equiv b \pmod{\Lambda}$ . □

We claim that  $\Phi$  is bijection. We claim that  $P_0 \in E(\Lambda)$  has a unique preimage under  $\Phi$ . If  $O$ , the unique preimage is  $0 \pmod{\Lambda}$ . Otherwise, let  $P_0 = (x_0, y_0) \in E(\Lambda)$ . Consider  $f(z) = \wp(z) - x_0$ , then  $f$  has 2 zeros mod  $\Lambda$ , let  $z_0$  be one of them (by Liouville's Theorem). Then  $f(z_0) = \wp(z_0) - x_0 = 0$ . Since  $\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3$  so  $\wp'(z_0)^2 = 4\wp(z_0)^3 - g_2\wp(z_0) - g_3 = 4x_0^3 - g_2x_0 - g_3 = y_0^2$  since  $P = (x_0, y_0) \in E(\Lambda)$  i.e.  $\wp'(z_0) = \pm y_0$ . We have two cases:

- (i) Case I:  $y_0 = 0$  then  $\wp'(z_0) = 0 = y_0$ .  $\Phi(z_0) = (\wp(z_0), \wp'(z_0)) = (x_0, y_0)$ . Suppose  $\Phi(z_1) = P_0$ . Since  $\wp'(z_0) = \wp'(z_1) = 0$ ,  $z_0, z_1 \in \{\frac{\omega_1}{2}, \frac{\omega_2}{2}, \frac{\omega_3}{2}\}$  by the Lemma. So  $\Phi(z_1), \Phi(z_0)$  agreeing in  $y$ -coordinate implies  $z_0, z_1 \in \{\frac{\omega_1}{2}, \frac{\omega_2}{2}, \frac{\omega_3}{2}\}$ .  $x_0 = \wp(z_0) = \wp(z_1)$ . Let  $z_0 = \frac{\omega_j}{2}, z_1 = \frac{\omega_k}{2}$ .  $x_0 = \wp(\frac{\omega_j}{2}) = \wp(\frac{\omega_k}{2})$  this contradicts to  $\{\frac{\omega_1}{2}, \frac{\omega_2}{2}, \frac{\omega_3}{2}\}$  are distinct.
- (ii) Case II:  $y_0 \neq 0$ .  $y_0 \neq 0 \implies \wp'(z_0) \neq 0$ . Let  $f(z) = \wp(z) - \wp(z_0)$ , then  $f(z_0) = 0$  and  $f'(z_0) = \wp'(z_0) \neq 0$ . So  $f$  has a simple zero at  $z_0$ . By Lemma,  $2z_0 \notin \Lambda$ . Also the zeros of  $f$  in  $P$  are  $\pm z_0 \pmod{\Lambda}$ . Remember that  $\wp'(z_0) = \pm y_0$ . Since  $\pm'$  is odd function,  $\wp'(-z_0) = \mp y_0$  so exactly one of  $z_0, -z_0$  give  $\wp(z_0) = y_0$ . And  $z_0, -z_0$  were the only candidates mod  $\Lambda$  for which  $\wp(z) = x_0$ .  $P_0$  has a unique preimage under  $\Phi$  i.e.  $\Phi$  is a bijection.

**Lemma 3.12.** Let  $A, B$  be abelian groups, let  $\varphi$  be a function. Suppose

- (i)  $f(0) = 0$ .
- (ii) for all  $a_1, a_2, a_3 \in A$ ,  $a_1 + a_2 + a_3 = 0 \implies \varphi(a_1) + \varphi(a_2) + \varphi(a_3) = 0$ .

Then  $\varphi$  is a homomorphism.

**Definition 3.13** ( $n$ -torsion).  $E(\mathbb{C})[n] = \{P \in E(\mathbb{Q}) : nP = O\}$  "order divide  $n$ ".

**Remark.**  $E(\mathbb{C})[n] \cong \mathbb{Z}/n\mathbb{Z} \oplus \mathbb{Z}/n\mathbb{Z}$ .

**Example.** Rational Points of Finite Order:

- (1) Points of order 2:  $y^2 = x^3 + ax^2 + bx + c$ . Correspond to  $(x, y)$ ,  $y = 0$ . Points in  $E(\mathbb{Q})$  of order 2:

$$\{(x, y) : y = 0, x \text{ a rational root of } x^3 + ax^2 + bx + c\}$$

Possible number of roots: 0, 1, 3. Then  $E(\mathbb{Q})[2] \cong 0, \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ .

- (2) Points of order 3:  $E(\mathbb{C})[3] \cong \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$ . We know that  $E(\mathbb{Q})[3]$  is a subgroup of  $E(\mathbb{C})[3]$  so options we have:

$$0, \quad \mathbb{Z}/3\mathbb{Z}, \quad \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$$

**Proposition 3.14.**  $E(\mathbb{R})[3] \not\cong \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$ .

**Proof.** Fix  $E : y^2 = x^3 + ax + b$ , we see that  $P = (x, y)$  has order 3 if and only if  $x$  is a root of

$$\Psi(x) = 3x^4 + 6ax^2 + 12bx - a^2$$

We claim  $\Psi$  has less than 4 roots: Suppose that  $\Psi$  has 4 real roots and we see that all roots are distinct. Then we must have  $a < 0$ . Let  $\lambda = \sqrt{-a} \in \mathbb{R}$ , consider  $\Psi(\lambda x)/\lambda^4 = 3x^4 - 6x^2 + (\frac{12b}{(-a)^{3/2}})x - 1$ . Thus, this equation must also have 4 real roots as well. As  $c$  varies, theoretically, the number of real roots of  $\Psi_c$  can change, this must occur at some  $c$  value for which  $\Psi_c$  has a double root. By computation,  $\Delta\Psi_c = -27(3c^2 - 6c)^2$  so  $\Delta\Psi_c = 0$  if and only if  $c = \pm \frac{8}{\sqrt{3}}$ . Then just check  $\Psi_0(x)$  has 2 real roots.  $\Psi_5(x), \Psi_{-5}(x)$  have 2 real roots so  $\Psi(x)$  has at most 2 real roots. Thus  $\Psi(x)$  has at most 2 real roots.  $\square$

$E(\mathbb{Q})[n]$  can be large, doesn't mean  $E(\mathbb{Q})[n]$  is necessarily large.

**Definition 3.15.**

$$E(\mathbb{Q})_{\text{tors}} = \{P \in E(\mathbb{Q}) : nP = O \text{ for some } n\} = \bigcup_{n=1}^{\infty} E(\mathbb{Q})[n]$$

called the torsion subgroup of  $E$ .

**Proposition 3.16.**  $E(\mathbb{Q})_{\text{tors}} \leq E(\mathbb{Q})$  is a subgroup.

**Proof.** Let  $P, Q \in E(\mathbb{Q})_{\text{tors}}$  then there exists  $n, m$  such that  $nP = O, mQ = O$  so

$$mn(P + Q) = mnP + nmQ = O$$

Other group axioms also follow easily.  $\square$

How big is  $E(\mathbb{Q})_{\text{tors}}$ ? In fact, finite! This is a consequence of

**Theorem 3.17 (Nagell-Lutz).** Let  $E : y^2 = x^3 + ax^2 + bx + c$ ,  $a, b, c \in \mathbb{Z}$  be an elliptic curve. Let  $P = (x, y) \in E(\mathbb{Q})_{\text{tors}}$ , other than  $O$ . Then

- (1)  $x, y \in \mathbb{Z}$ .
- (2) Either  $y = 0$  or  $y^2 \mid \Delta$  where  $\Delta$  is the discriminant of  $x^3 + ax^2 + bx + c$ .

**Remarks.**

- (1) NOT if and only if:
$$(x, y) \in \mathbb{Z}^2 \not\Rightarrow (x, y) \in E(\mathbb{Q})_{\text{tors}}$$
- (2) Finitely many choices of  $y$ , each  $y$  gives  $\leq 3$   $x$ -values implies  $E(\mathbb{Q})_{\text{tors}}$  is finite.

Algorithm for computing  $E(\mathbb{Q})_{\text{tors}}$ :

(Step 0): Put  $E$  into the form

$$E : y^2 = x^3 + ax^2 + bx + c, \quad a, b, c \in \mathbb{Z}$$

(Step 1): Compute  $\Delta$

$$\Delta = a^2b^2 - 4b^3 - 4a^3c + 18abc - 27c^2$$

List options for  $y$ ,  $y \in \{0\} \cup \{y : y^2 \mid \Delta\}$ .

(Step 2): For each  $y$  in Step 1, consider

$$x^3 + ax^2 + bx + c - y^2 = 0$$

Find rational roots.

(Step 3): Which  $(x, y) \in \mathbb{Z}^2$  are actually torsion? Let  $P_1, \dots, P_n$  be the candidates from Step 2. For each, compute  $P, 2P, 4P, 8P, \dots$ . Two options:

- (1) Starting repeating:  $P \in E(\mathbb{Q})_{\text{tors}}$ .
- (2) Doesn't repeat: then find some  $2^k P \notin \{P_1, \dots, P_n\}$ . This implies  $P$  was not torsion,  $P \notin E(\mathbb{Q})_{\text{tors}}$ .

**Example.**  $y^2 = x^3 - 4x^2 - x + 4$ . Compute  $E(\mathbb{Q})_{\text{tors}}$ .

(Step 1):  $\Delta = 16 + 4 + 1024 + 18 \cdot 16 - 27 \cdot 16 = 900$ . Thus,  $y \in \{0, \pm 2, \pm 3, \pm 5, \pm 6, \pm 10, \pm 15, \dots, \pm 30\}$ .

(Step 2):  $y = 0$ , roots of  $x^3 - 4x^2 - x + 4$  implies  $x \mid 4$ ,  $x \in \{\pm 1, \pm 2, \pm 4\}$  then  $x = \pm 1, 4$  as roots.

$y = \pm 1$ : Roots of  $x^3 - 4x^2 - x + 3$  but one can check there's no rational root.

$y = \pm 2$ : Roots of  $x^3 - 4x^2 - x$  and  $x = 0$  works. Similarly, check that no other  $y$  leads to  $(x, y) \in \mathbb{Z}^2$ . Candidate torsion points are

$$\{0, (-1, 0), (1, 0), (4, 0), (0, 2), (0, -2)\}$$

(Step 3): Know  $0, (-1, 0), (1, 0), (4, 0)$  are all torsion points of order 2. Check  $(0, 2)$ : slope at  $(0, 2)$  is

$$\frac{f'(x)}{2y} = \frac{3x^2 - 8x - 1}{2y} = -\frac{1}{4}$$

Tangent line is  $y - 2 = -\frac{1}{4}(x - 0)$ .

$$\begin{aligned} \left(-\frac{1}{4}x + 2\right)^2 &= x^3 - 4x^2 - x + 4 \\ x^3 - \frac{65}{16}x^2 &= 0 \implies x^2\left(x - \frac{65}{16}\right) = 0 \end{aligned}$$

Thus,  $2P$  has  $x$ -coordinate  $\frac{65}{16} \notin \mathbb{Z}$ . This means  $2P$  is not torsion point i.e.  $P$  is not torsion point so  $(0, 2) \notin E(\mathbb{Q})_{\text{tors}}$ .

We conclude that

$$E(\mathbb{Q})_{\text{tors}} = \{0, (-1, 0), (1, 0), (4, 0)\} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}.$$

**Theorem 3.18 (Mazur's Theorem).** Let  $E : y^2 = x^3 + ax^2 + bx + c$ ,  $a, b, c \in \mathbb{Z}$ . Then  $E(\mathbb{Q})_{\text{tors}}$  is isomorphic to one of

- (1)  $\mathbb{Z}/n\mathbb{Z}$ .  $n = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12$ .
- (2)  $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2m\mathbb{Z}$ ,  $m = 1, 2, 3, 4$ .

15 options in total.

Let's prove the easy part of Nagell-Lutz:

**Proposition 3.19.** Let  $E : y^2 = x^3 + ax^2 + bx + c$ ,  $a, b, c \in \mathbb{Z}$ . Let  $P = (x, y)$ ,  $y \neq 0$ . If  $2P$  has integer coordinates, then  $y^2 \mid \Delta$ .

**Proof.** Let  $P = (x_P, y_P)$ ,  $2P = (x_{2P}, y_{2P})$ . Tangent line to  $E$  at  $P$  is

$$y - y_P = \lambda(x - x_P) \quad \lambda = \frac{f'(x_P)}{2y_P}$$

$$\begin{aligned} \implies y &= \lambda(x - x_P) + y_P \\ \implies (\lambda(x - x_P) + y_P)^2 &= x^3 + ax^2 + bx + c \end{aligned}$$

And we see that the sum of the roots are  $\lambda^2 - a$ .  $x_{2P} + 2x_P = \lambda^2 - a$ ,  $x_{2P}, x_P, a \in \mathbb{Z} \implies \lambda^2 \in \mathbb{Z}$  so  $\lambda \in \mathbb{Z}$ .  $2y_P \mid f'(x_P)$ . Let  $g(x) = f(x + x_P)$ . Since  $f(x) \in \mathbb{Z}[x]$ ,  $g(x) \in \mathbb{Z}[x]$  say  $g(x) = x^3 + Ax^2 + Bx + C$ ,  $A, B, C \in \mathbb{Z}$ .  $C = g(0) = f(x_P) = y_P^2$ .  $B = g'(0) = f'(x_P) = 2\lambda y_P$ . Since we have  $\Delta(g) = \Delta(f) = A^2B^2 - 4B^3 - 4A^3C + 18ABC - 27C^2$  and each term at right is divisible by  $y_P^2$  so  $y_P^2 \mid \Delta$ .  $\square$

**Remark.** Suppose we know that  $(x, y) \in E(\mathbb{Q})_{\text{tors}}$  had  $x, y \in \mathbb{Z}$ . Let  $P = (x, y) \in E(\mathbb{Q})_{\text{tors}}$ ,  $2P \in E(\mathbb{Q})_{\text{tors}}$ . By N-L,  $2P$  has integer coordinates so  $y^2 \mid \Delta$  or  $y = 0$ .

**Theorem 3.20 (Nagell-Lutz., Strong Form).** Let  $E : y^2 = x^3 + ax^2 + bx + c$ ,  $a, b, c \in \mathbb{Z}$ . Let  $p = (x, y) \in E(\mathbb{Q})_{\text{tors}}$ . Then

- (a)  $x, y \in \mathbb{Z}$ .
- (b)  $y = 0$  or  $y^2 \mid \Delta$  where  $\Delta$  is the discriminant of  $x^3 + ax^2 + bx + c$ .

**Definition 3.21.** Order at  $p$

$$x = p^v \frac{m}{n} \in \mathbb{Q}, \quad p \nmid m, n \Leftrightarrow \text{ord}_p x = v$$

**Proposition 3.22.** If  $(x, y) \in E(\mathbb{Q})$  and  $\text{ord}_p(x) < 0$  or  $\text{ord}_p(y) < 0$ . Then  $\text{ord}_p(x) = -2v$  and  $\text{ord}_p(y) = -3v$  for some integer  $v > 0$ .

**Definition 3.23.**  $E(p^v) = \{(x, y) \in E(\mathbb{Q}) : \text{ord}_p(x) \leq -2v, \text{ord}_p(y) \leq -3v\} \cup \{O\}$ .

**Remark.** Note that  $E(\mathbb{Q}) \supseteq E(p) \supseteq E(p^2) \supseteq \dots$ . In fact this is subgroup inclusions.

To prove this, we use change of coordinates

$$[X : Y : Z] = [S : T : U] = [Z : X : Y]$$

Take an equation

$$\begin{aligned} Y^2Z &= X^3 + aX^2Z + bXZ^2 + cZ^3 \\ U^2S &= T^3 + aT^2S + bTS^2 + cS^3 \end{aligned}$$

impose affine coordinates  $s = \frac{S}{U}$  and  $t = \frac{T}{U}$ .

$$\implies s = t^3 + at^2 + bts^2 + cs^3$$

Observations

- (1)  $(x, y) \mapsto [x : y : 1] \mapsto [1 : x : y]$ . If  $y \neq 0$ ,  $s = \frac{1}{y}$ ,  $t = \frac{x}{y}$  then  $(x, y) \mapsto (t, s) = (\frac{x}{y}, \frac{1}{y})$ .
- (2)  $O = [0 : 1 : 0] \mapsto [0 : 0 : 1]$  so  $(t, s) = (0, 0)$ .
- (3)  $(x, y)$  with  $y = 0$  we get mapped outside the  $(t, s)$  affine plane.
- (4)  $(x, y) \in E(p^v)$  and  $x = p^{-2(v+k)} \frac{m}{n}$ ,  $y = p^{-3(v+k)} \frac{q}{r}$ . Then

$$t = p^{v+k} \frac{mq}{nr}, \quad s = p^{3(v+k)} \frac{r}{q}$$

so  $t \in p^v \mathbb{Z}_{(p)}$ ,  $s \in p^{3v} \mathbb{Z}_{(p)}$ . So  $(x, y) \in E(p^v) \Leftrightarrow t \in p^v \mathbb{Z}_{(p)}$ ,  $s \in p^{3v} \mathbb{Z}_{(p)}$ .

Upshot:  $(t, s)$  coordinates make it easier to take advantage of closure properties coming from  $\mathbb{Z}_{(p)}$ .

- (5) Projective linear transformations map lines to lines. Since the group law is defined via intersections of lines. We can add points the same geometric way.

**Lemma 3.24.** Let  $P_1 = (t_1, s_1)$ ,  $P_2 = (t_2, s_2)$ .  $P_1, P_2 \in E(p^v)$ . Suppose that  $t_1 \neq t_2$ . Then

$$\frac{s_2 - s_1}{t_2 - t_1} = \alpha \in p^{2v} \mathbb{Z}_{(p)}$$

**Proof.**  $P_1, P_2 \in E(\mathbb{Q})$ :

$$\begin{aligned} s_1 &= t_1^3 + at_1^2 s_1 + bt_1 s_1^2 + cs_1^3 \\ s_2 &= t_2^3 + at_2^2 s_2 + bt_2 s_2^2 + cs_2^3 \end{aligned}$$

Subtract:

$$\begin{aligned} s_2 - s_1 &= t_2^3 - t_1^3 + a(t_2^2 s_2 - t_1^2 s_1) + b(t_2 s_2^2 - t_1 s_1^2) + c(s_2^3 - s_1^3) \\ &= t_2^3 - t_1^3 + as_2(t_2^2 - t_1^2) + at_1^2(s_2 - s_1) + bs_2^2(t_2 - t_1) + bt_1(s_2^2 - s_1^2) + c(s_2^3 - s_1^3) \\ &= (t_2 - t_1)(t_2^2 + t_1 t_2 + t_1^2 + as_2(t_1 + t_2) + bs_2^2) + (s_2 - s_1)(at_1^2 + bt_1(s_1 + s_2) + c(s_1^2 + s_1 s_2 + s_2^2)) \end{aligned}$$

Solve for  $\alpha = \frac{s_2 - s_1}{t_2 - t_1}$ ,

$$\alpha = \frac{t_2^2 + t_1 t_2 + t_1^2 + as_2(t_1 + t_2) + bs_2^2}{1 - at_1^2 - bt_1(s_1 + s_2) - c(s_1^2 + s_1 s_2 + s_2^2)}$$

Suppose  $P_1, P_2 \in E(p^v)$ . Then  $\text{ord}_p(\text{numerator}) = 2v$  and  $\text{ord}_p(\text{denominator}) = 0$ . Thus,  $\text{ord}_p(\alpha) \geq 2v - 0 = 2v$ .  $\square$

**Proposition 3.25.** Let  $v \geq 1$ .  $E(p^v)$  is a subgroup of  $E(\mathbb{Q})$ .

**Proof.** Let  $P_1, P_2 \in E(p^v)$ . In  $(t, s)$ -coordinates,  $P_1 = (t_1, s_1)$ ,  $P_2 = (t_2, s_2)$ .  $P_1 + P_2 = (t_3, s_3)$ . Given that  $t_1, t_2 \in p^v \mathbb{Z}_{(p)}$  and  $s_1, s_2 \in p^{3v} \mathbb{Z}_{(p)}$ . We want to show that  $t_3 \in p^v \mathbb{Z}_{(p)}$  and this implies  $s_3 \in p^{3v} \mathbb{Z}_{(p)}$ . Assume  $P_1 \neq P_2$  then

- (1) If  $t_1 = t_2$ , the line joining  $P_1$  and  $P_2$  is vertical.  $P_1 * P_2 = (t_1, s_0)$  for some  $s_0$ . Then  $P_1 + P_2 = (-t_1, -s_0) = (t_3, s_3)$  so  $t_3 = -t_1 \in p^v \mathbb{Z}_{(p)}$ .
- (2) If  $t_1 \neq t_2$ . Let  $\alpha = \frac{s_2 - s_1}{t_2 - t_1}$ . The line joining  $P_1$  and  $P_2$  has equation  $s = \alpha t + \beta$ .  $\beta = s - \alpha t$ . This intersects  $t$  when

$$\begin{aligned} s &= t^3 + at^2 s + bt s^2 + cs^3 \\ \implies \alpha t + \beta &= t^3 + at^2(\alpha t + \beta) + bt(\alpha t + \beta)^2 + c(\alpha t + \beta)^3 \end{aligned}$$

Simplify

$$(1 + a\alpha + b\alpha^2 + c\alpha^3)t^3 + (a\beta + 2b\alpha\beta + 3c\alpha^2\beta)t^2 + \dots = 0$$

This has roots  $t_1, t_2, -t_3$ . By sum of roots we know that

$$t_1 + t_2 - t_3 = -\frac{a\beta + 2b\alpha\beta + 3c\alpha^2\beta}{1 + a\alpha + b\alpha^2 + c\alpha^3}$$

We know that  $t_1, t_2 \in p^v \mathbb{Z}_{(p)}$  and  $\alpha \in p^{2v} \mathbb{Z}_{(p)}$  so  $\beta = s_1 - \alpha t_1 \in p^{3v} \mathbb{Z}_{(p)}$ . Then  $\text{ord}_p(\text{numerator}) = 3v$  and  $\text{ord}_p(\text{denominator}) = 0$  i.e.  $t_1 + t_2 - t_3 \in p^{3v} \mathbb{Z}_{(p)}$ . Thus,  $t_3 \in p^v \mathbb{Z}_{(p)}$ .  $\square$

**Remark.** We proved more than we needed:  $t_1 + t_2 - t_3 \in p^{3v} \mathbb{Z}_{(p)}$  v.s.  $p^v \mathbb{Z}_{(p)}$ . Consider the function  $P \mapsto t(P)$ , the  $t$ -coordinate of  $P$  as a map  $E(p^v) \rightarrow p^v \mathbb{Z}_{(p)} / p^{3v} \mathbb{Z}_{(p)}$ . Since  $t(P_1) + t(P_2) \equiv t(P_1 + P_2) \pmod{p^{3v} \mathbb{Z}_{(p)}}$ . The map  $P \rightarrow t(P)$  is a homomorphism. The kernel is  $E(p^{3v})$  so

$$E(p^v) / E(p^{3v}) \rightarrow p^v \mathbb{Z}_{(p)} / p^{3v} \mathbb{Z}_{(p)}$$

is an injective homomorphism.

**Theorem 3.26** (Nagell-Lutz Theorem).  $P \in E(\mathbb{Q})_{\text{tors}}$ , if  $p$  is prime then

$$P \notin E(p)$$

**Proof.** Take  $P \in E(\mathbb{Q})_{\text{tors}}$  of order  $m$ . Suppose  $P \in E(p)$ ,  $E(p) \supseteq E(p^2) \supseteq \dots$ . There exists some  $v \geq 1$ ,  $P \in E(p^v)$ ,  $P \notin E(p^{v+1})$ .

- (1) Suppose that  $p \nmid m$ , then  $t(mP) = t(O) = t(0, 0) = 0 \equiv mt(P) \pmod{p^{3v}\mathbb{Z}_{(p)}}$ . Then  $mt(P) \in p^{3v}\mathbb{Z}_{(p)}$  so  $t(p) \in p^{3v}\mathbb{Z}_{(3p)}$ . Thus implies  $P \in E(p^{3v})$ . Since  $3v \geq v+1$  we get a contradiction.
- (2) Suppose  $p \mid m$ , say  $m = pm_0$ , write  $P' = m_0P$ . So  $P'$  has order  $p$  in  $E(\mathbb{Q})$ . Choose  $v'$  so that  $P' \in E(p^{v'}) \setminus E(p^{v'+1})$ . Then

$$t(pP') = t(O) = 0 \equiv pt(P') \pmod{p^{3v'}\mathbb{Z}_{(p)}}$$

so  $pt(P') \in p^{3v'}\mathbb{Z}_{(p)}$  i.e.  $t(P') \in p^{3v'-1}\mathbb{Z}_{(p)}$ . Thus,  $P' \in E(p^{3v'-1})$  and it's clear that  $3v' - 1 \geq v' + 1$  so we get a contradiction.  $\square$

## 4 Elliptic Curves over Finite Fields

Given  $P = [X : Y : Z] \in \mathbb{P}^2(\mathbb{Q})$ , we can scale to assume  $X, Y, Z \in \mathbb{Z}$  and so that  $\gcd(X, Y, Z) = 1$ . In particular, given a prime  $p$ ,  $p$  does not divide  $X, Y, Z$  simultaneously. Then  $[\bar{X}, \bar{Y}, \bar{Z}] \neq [0 : 0 : 0]$  be reduction mod  $p$ . So  $[\bar{X}, \bar{Y}, \bar{Z}]$  is an element of  $\mathbb{P}^2(\mathbb{F}_p)$  where  $\mathbb{F}_p$  is finite field with  $p$  elements. We obtain a map  $\mathbb{P}^2(\mathbb{Q}) \rightarrow \mathbb{P}^2(\mathbb{F}_p)$  called reduction mod  $p$ .

**Definition 4.1.**  $[X : Y : Z] \in \mathbb{P}^2(\mathbb{Q})$  is called normalized if  $X, Y, Z \in \mathbb{Z}$  and  $\gcd(X, Y, Z) = 1$ .

**Remark.**  $[X, Y, Z]$  has 2 normalizations namely  $[X : Y : Z]$  and  $[-X : -Y : -Z]$ . Since  $[\bar{X}, \bar{Y}, \bar{Z}] = [-\bar{X}, -\bar{Y}, -\bar{Z}] \in \mathbb{P}^2(\mathbb{Q})$  reduction mod  $p$  is well-defined.

**Remark.** We don't have a map between affine spaces,  $\mathbb{A}^2(\mathbb{Q}) \rightarrow \mathbb{A}^2(\mathbb{F}_p)$ .

**Example.**  $(1, \frac{1}{2}) \rightarrow [1 : \frac{1}{2} : 1] \rightarrow [2 : 1 : 2]$  normalized and

$$[2 : 1 : 2] \rightarrow [0, 1, 0] \text{ reduce mod } 2$$

does not lie in  $\mathbb{A}^2(\mathbb{F}_p)$ .

Let  $F(X, Y, Z) = 0$  be some algebraic curve  $\mathbb{P}^2(\mathbb{Q})$ . (so  $F$  has  $\mathbb{Q}$  coefficients). As before, scale  $F$  so that the coefficients are integers, and their  $\gcd = 1$ . Then reduce all coefficients mod  $p$  and consider the curve  $\bar{F}(X, Y, Z) = \sum \bar{c}_j X^{e_j} Y^{f_j} Z^{g_j}$ . Consider

$$\bar{F}(X, Y, Z) \equiv 0 \pmod{p}$$

Points on  $\bar{F}$  are  $[X : Y : Z] \in \mathbb{P}^2(\mathbb{F}_p)$  satisfying  $\bar{F}(X, Y, Z) \equiv 0 \pmod{p}$ . " $\bar{F}$  is a projective curve over  $\mathbb{F}_p$ ". Elliptic Curves:

$$E : Y^2Z = X^3 + aX^2Z + bXZ^2 + cZ^3$$

Define the reduction of  $E$  mod  $p$ ,  $\bar{E}$  as the curve

$$Y^2Z = X^3 + \bar{a}X^2Z + \bar{b}XZ^2 + \bar{c}Z^3$$

again viewed as a projective curve. Points on  $\bar{E}$ :

1. If  $Z \equiv 0 \pmod{p}$  then  $X^3 \equiv 0 \pmod{p}$  so  $X \equiv 0 \pmod{p}$ . Get the point

$$[X : Y : Z] = [0 : Y : 0] = [0 : 1 : 0] \in \mathbb{P}^2(\mathbb{F}_p).$$

2. Otherwise,  $Z \not\equiv 0 \pmod{p}$ , and we scale  $[X : Y : Z]$  by  $Z^{-1} \pmod{p}$ . It suffices to study  $[X : Y : 1]$  so we go back to affine spaces. Consider  $(x, y) \in \mathbb{F}_p^2$  for which

$$y^2 \equiv x^3 + \bar{a}x^2 + \bar{b}x + \bar{c} \pmod{p}$$

These curves define elliptic curves over  $\mathbb{F}_p$  provided they are non-singular. What do tangents look like over  $\mathbb{P}^2(\mathbb{F}_p)$ ? Over  $\mathbb{C}$ , non-singular iff  $\Delta \neq 0$ . Over  $\mathbb{F}_p$ , non-singular iff  $\Delta \not\equiv 0 \pmod{p}$  i.e.  $p \nmid \Delta$ .

**Definition 4.2.** Let  $E$  be an elliptic curve over  $\mathbb{Q}$ . Let  $\bar{E}$  be the reduction mod  $p$ . Then  $\bar{E}$  is an elliptic curve if  $p \nmid \Delta$ . Let  $\bar{E}(\mathbb{F}_p)$  denote the set of  $\mathbb{F}_p$  points on  $\bar{E}$ .

**Proposition 4.3.**  $\bar{E}(\mathbb{F}_p)$  is an abelian group.

**Proof.** We define a group law on  $\mathbb{E}(\mathbb{C})$  geometrically. But when in Weierstrass form, we also explicit formulas:  $P_1 = (x_1, y_1), P_2 = (x_2, y_2)$ . Let

$$\lambda = \begin{cases} \frac{y_2 - y_1}{x_2 - x_1} & P_1 \neq P_2 \\ \frac{3x_1^2 + 2ax_1 + b}{2y_1} & P_1 = P_2 \end{cases}$$

and we omit the cases when  $P_1 = -P_2$ . Tangent/secant line at  $P_1, P_2$  is  $y = \lambda x + \nu$  where  $\nu = y_1 - \lambda x_1 = y_2 - \lambda x_2$ . By sum of roots, we have

$$x_3 = \lambda^2 - a - x_1 - x_2, \quad y_3 = -(\lambda x_3 + \nu)$$

To make a group law on  $\bar{E}(\mathbb{F}_p)$ , we take these explicit formulas and interpret mod  $p$  so denominators are mod  $p$  inverses.

Caution: We used Bézout, which only works over algebraically closed fields.  $\mathbb{F}_p$  is not algebraically closed, so work over  $\overline{\mathbb{F}}_p$ .  $\square$

**Proposition 4.4.**  $E(\mathbb{Q}) \hookrightarrow \overline{E}(\mathbb{F}_p)$  is a homomorphism.

**Proof.** To show a homomorphism, it's enough to show that

$$P_1 + P_2 + P_3 = O \implies \overline{P}_1 + \overline{P}_2 + \overline{P}_3 = O$$

Cases:

1. If  $P_j = 0$  then it's trivial.
2. If two of the  $P_j = O$ , then all 3 are  $O$ .
3. If  $P_1 = O$ , let  $P_2 = (x, y)$  then  $P_3 = (x, -y)$ . If we assume  $P_2, P_3 \in E(\mathbb{Q})_{\text{tors}}$  then Nagell-Lutz implies  $x, y \in \mathbb{Z}$ . Thus,  $P_1 = [X : Y : 1]$  is a normalized point so  $\overline{P}_2 = [\overline{X} : \overline{Y} : 1], \overline{P}_3 = [\overline{X}, -\overline{Y} : 1]$  so  $\overline{P}_2 = -\overline{P}_3$ .
4. Then  $P_1, P_2, P_3$  lie on a line  $L : y = \lambda x + \nu$ . We know that  $x_1 + x_2 + x_3 = \lambda^2 - a$ . Since  $x_1, x_2, x_3, a \in \mathbb{Z}$  we know that  $\lambda \in \mathbb{Z}$ . Also  $\nu = y_1 - \lambda x_1 \in \mathbb{Z}$  so  $\overline{L} : Y = \lambda X + \nu Z$ .

$$x^3 + ax^2 + bx + c - (\lambda x + \nu)^2 = (x - x_1)(x - x_2)(x - x_3)$$

Reduce this mod  $p$ , we have  $\overline{E}$  and  $\overline{L}$  intersect at  $\overline{x}_1, \overline{x}_2, \overline{x}_3$ . Thus,  $\overline{P}_1, \overline{P}_2, \overline{P}_3$  lie on  $\overline{L}$  so  $\overline{P}_1 + \overline{P}_2 + \overline{P}_3 = O$ .  $\square$

**Proposition 4.5.**  $E(\mathbb{Q})_{\text{tors}} \hookrightarrow \overline{E}(\mathbb{F}_p)$  is injective. In particular,  $|E(\mathbb{Q})_{\text{tors}}|$  divides  $|\overline{E}(\mathbb{F}_p)|$ .

**Proof.**  $P = (x, y) \in E(\mathbb{Q})_{\text{tors}}$ ,  $P \neq O$ . Then, by Nagell-Lutz,  $x, y \in \mathbb{Z}$  so  $[X : Y : 1]$  is normalized.  $\overline{P} = [\overline{x} : \overline{y} : 1] \in \mathbb{P}^2(\mathbb{F}_p)$ .  $\overline{P} \neq [0 : 1 : 0]$  so  $\overline{P}$  does not lie in the kernel i.e.  $\ker = \{O\}$  so we inject. In addition,  $E(\mathbb{Q})_{\text{tors}} \cong$  subgroups of  $\overline{E}(\mathbb{F}_p)$  so  $|E(\mathbb{Q})_{\text{tors}}|$  divides  $|\overline{E}(\mathbb{F}_p)|$ .  $\square$

**Example.** Let  $E : y^2 = x^3 + 5x - 10$ , what is  $E(\mathbb{Q})_{\text{tors}}$ ?

1. Compute

$$\Delta = -4 \cdot 5^3 - 27 \cdot 10^2 = -2^7 \cdot 5^2$$

Since  $3 \nmid \Delta$ , we will think about reduction mod 3. Then  $\overline{E}(\mathbb{F}_3)$  contains  $O$  and contains affine points.

$$\begin{array}{c|c|c|c} x & 0 & 1 & 2 \\ \hline x^3 + 5x - 10 \pmod{3} & 2 & 2 & 2 \end{array}$$

2 is a quadratic non-residue mod 3 so  $y^2 \equiv 2 \pmod{3}$  has no solutions. Thus,  $\overline{E}(\mathbb{F}_p) = \{O\}$  which implies  $E(\mathbb{Q})_{\text{tors}} = \{O\}$  by the proposition above.

## 5 The Mordell-Weil Theorem

**Theorem 5.1 (Mordell-Weil).** Let  $E$  be an elliptic curve over  $\mathbb{Q}$ . Then  $E(\mathbb{Q})$  is a finitely generated group. In fact,  $E(\mathbb{Q})$  is sometimes called the Mordell-Weil group.

**Corollary 5.2.**  $\mathbb{E}(\mathbb{Q})$  is isomorphic to  $\mathbb{E}(\mathbb{Q})_{\text{tors}} \oplus \mathbb{Z}^{r_E}$  in which  $r_E$  is called the rank of  $E$ .

Two steps to prove Mordell-Weil Theorem

1. Let  $2E(\mathbb{Q}) = \{2P : P \in E(\mathbb{Q})\}$ . This is a subgroup of  $\mathbb{E}(\mathbb{Q})$ . We would show that  $E(\mathbb{Q})/2E(\mathbb{Q})$  is a finite group.
2. Prove that the above tells us the Mordell-Weil.

**Definition 5.3.** Let  $x \in \mathbb{Q}$ ,  $x = \frac{m}{n}$  in lowest terms. The height of  $x$ ,  $H(x)$  is  $H(x) = \max\{|m|, |n|\}$ . In particular, the height of infinity is  $H(\infty) = H(\frac{1}{0}) = 1$ .

**Example.**  $H(\frac{5}{2}) = 5$ .  $H(\frac{9999}{10000}) = 10000$ .

**Lemma 5.4.** Let  $B > 0$ . Then  $\{x \in \mathbb{Q} : H(x) < B\}$  is finite.

**Proof.**  $H(x) < B$  implies  $x = \frac{m}{n}$  and  $|m| \leq B$  and  $|n| \leq B$ . We have at most  $(2B+1)^2$  options for  $(m, n)$  which means we have finitely many options for  $x$ .  $\square$

**Proposition 5.5.** Let  $x, y \in \mathbb{Q}$  then

1.  $H(x^{-1}) = H(x)$ .

2. For  $d \in \mathbb{Z}$ ,  $d \geq 1$ ,  $H(x^d) = H(x)^d$ .
3.  $H(xy) \leq H(x)H(y)$ .

**Proof.**

1. Trivial.
2. Let  $x = \frac{m}{n}$  with  $(m, n) = 1$  then  $H((\frac{m}{n})^d) = \max\{|m|^d, |n|^d\} = H(\frac{m}{n})^d$ .
3. Let  $y = \frac{p}{q}$  and  $x = \frac{m}{n}$  then

$$H(xy) = H(\frac{mp}{nq})$$

Note that  $\gcd(mp, nq)$  might be large. But it's clear that any cancellation would actually reduce the height. As an upper bound.

$$H(\frac{mp}{nq}) \leq \max\{|mp|, |nq|\} \leq \max\{|m|, |n|\} \cdot \max\{|p|, |q|\} = H(x)H(y).$$

□

**Proposition 5.6.** Let  $p(x), q(x) \in \mathbb{Z}[x]$ . Let  $d = \max\{\deg p, \deg q\}$ , then there exists a constant  $K$  depending only on  $p$  and  $q$  for which

$$H(\frac{p(x)}{q(x)}) \leq KH(x)^d$$

**Proof.** Let

$$\begin{aligned} p(x) &= a_d x^d + a_{d-1} x^{d-1} + \cdots + a_1 x + a_0 \\ q(x) &= b_d x^d + b_{d-1} x^{d-1} + \cdots + b_1 x + b_0 \end{aligned}$$

Let  $x = \frac{m}{n}$ ,  $(m, n) = 1$  then

$$\begin{aligned} H(\frac{p(x)}{q(x)}) &= H(\frac{a_d x^d + a_{d-1} x^{d-1} + \cdots + a_1 x + a_0}{b_d x^d + b_{d-1} x^{d-1} + \cdots + b_1 x + b_0}) \\ &= H(\frac{a_d (\frac{m}{n})^d + a_{d-1} (\frac{m}{n})^{d-1} + \cdots + a_1 (\frac{m}{n}) + a_0}{b_d (\frac{m}{n})^d + b_{d-1} (\frac{m}{n})^{d-1} + \cdots + b_1 (\frac{m}{n}) + b_0}) \\ &= H(\frac{a_d m^d + a_{d-1} m^{d-1} n + \cdots + m n^{d-1} a_1 + n^d a_0}{b_d m^d + b_{d-1} m^{d-1} n + \cdots + m n^{d-1} b_1 + n^d b_0}) \end{aligned}$$

May not be in lowest terms, and we get an upper bound

$$\begin{aligned} H(\frac{p(x)}{q(x)}) &\leq \max\{|a_d m^d + a_{d-1} m^{d-1} n + \cdots + m n^{d-1} a_1 + n^d a_0|, |b_d m^d + b_{d-1} m^{d-1} n + \cdots + m n^{d-1} b_1 + n^d b_0|\} \\ &\leq \max\{|a_d| |m|^d + |a_{d-1}| |m|^{d-1} n + \cdots + |m| |n|^{d-1} |a_1| + |n|^d |a_0|, |b_d| |m|^d + |b_{d-1}| |m|^{d-1} n + \cdots + |m| |n|^{d-1} |b_1| + |n|^d |b_0|\} \end{aligned}$$

Let  $C = \max\{|a_d|, \dots, |a_0|, |b_d|, \dots, |b_0|\}$  then we have

$$H(\frac{p(x)}{q(x)}) \leq \max\{|m|^d + |m|^{d-1} |n| + \cdots + |n|^d, |m|^d + \cdots + |n|^d\}$$

Each term, we have  $|m|^{d-\alpha} |n|^\alpha \leq \max\{|m|, |n|\}^d$ . Then

$$H(\frac{p(x)}{q(x)}) \leq C(d+1) \max\{|m|, |n|\}^d = C(d+1)H(x)^d$$

Let  $K = C(d+1)$  then we are done. □

What about the lower bound? Define  $P(m, n) = n^d p(\frac{m}{n})$  and  $Q(m, n) = n^d q(\frac{m}{n})$ . Then  $P(m, n), Q(m, n) \in \mathbb{Z}[m, n]$  and

$$H(\frac{p(x)}{q(x)}) = H(\frac{P(m, n)}{Q(m, n)})$$

In upper bound, we ignored cancellation. For lower bound, we would have issues if  $\text{ord}(P(m, n), Q(m, n))$  were large.

**Lemma 5.7.** Let  $p(x), q(x) \in \mathbb{Z}[x]$ . Assume that  $p, q$  have no common roots in  $\mathbb{C}$ . Then there exists  $F_1(x, y), F_2(x, y), F_3(x, y), F_4(x, y) \in \mathbb{Z}[x, y]$  and  $A, B \in \mathbb{Z}$ ,  $A, B \geq 1$  for which

$$F_1(m, n)P(m, n) + F_2(m, n)Q(m, n) = Am^{D+d}$$

$$F_3(m, n)P(m, n) + F_4(m, n)Q(m, n) = Bn^{D+d}$$

where  $d = \max\{\deg p, \deg q\}$  and  $D = \max\{\deg F_j\}$ . Thus, if  $\gcd(m, n) = 1$ , then  $\gcd(P(m, n), Q(m, n))$  divides  $Am^{D+d}$  and  $Bn^{D+d}$ , so it divides  $\text{lcm}(A, B)$ .

**Proof.** Since  $p, q$  has no common roots in  $\mathbb{C}$ , they are coprime in  $\mathbb{Q}[x]$ . By the Euclidean algorithm, there exist  $r(x), s(x) \in \mathbb{Q}[x]$  so that  $p(x)r(x) + q(x)s(x) = 1 \in \mathbb{Q}[x]$ . By clearing denominators from  $r(x), s(x)$ , we can assume  $r(x), s(x) \in \mathbb{Z}[x]$ , but now  $p(x)r(x) + q(x)s(x) = B$  for some  $B \in \mathbb{Z}$ .

Define  $F_3(m, n) = n^D r(\frac{m}{n})$ ,  $F_4(m, n) = n^D s(\frac{m}{n})$ .  $F_3(m, n), F_4(m, n) \in \mathbb{Z}[m, n]$ . Then

$$\begin{aligned} F_3(m, n)P(m, n) + F_4(m, n)Q(m, n) &= n^D r(\frac{m}{n})n^d p(\frac{m}{n}) + n^D s(\frac{m}{n})n^d q(\frac{m}{n}) \\ &= n^{D+d}(r(x)p(x) + s(x)q(x)) = Bn^{D+d} \end{aligned}$$

Next, let  $\tilde{p}(x) = x^d p(\frac{1}{x})$  and  $\tilde{q}(x) = x^d q(\frac{1}{x})$ . Roots of  $\tilde{p}$  and  $\tilde{q}$  are inverses of roots of  $p, q$  respectively. We see that  $\tilde{p}$  and  $\tilde{q}$  are coprime in  $\mathbb{Q}[x]$ . As before, there exists  $\tilde{r}, \tilde{s} \in \mathbb{Z}[x]$  so that

$$\tilde{p}(x)\tilde{r}(x) + \tilde{q}(x)\tilde{s}(x) = A \in \mathbb{Z}[x]$$

Define  $F_1(m, n) = m^D \tilde{r}(\frac{n}{m})$  and  $F_2(m, n) = m^D \tilde{s}(\frac{n}{m})$ .  $P(m, n) = n^d p(\frac{m}{n}) = m^d \tilde{p}(\frac{n}{m})$  and  $Q(m, n) = n^d q(\frac{m}{n}) = m^d \tilde{q}(\frac{n}{m})$ . Then

$$\begin{aligned} F_1(m, n)P(m, n) + F_2(m, n)Q(m, n) &= m^D \tilde{r}(\frac{n}{m})m^d \tilde{p}(\frac{n}{m}) + m^D \tilde{s}(\frac{n}{m})m^d \tilde{q}(\frac{n}{m}) \\ &= m^{D+d}(\tilde{r}(\frac{1}{x})\tilde{p}(\frac{1}{x}) + \tilde{s}(\frac{1}{x})\tilde{q}(\frac{1}{x})) \\ &= Am^{D+d} \end{aligned}$$

Assume  $\gcd(m, n) = 1$  then  $\gcd(P(m, n), Q(m, n))$  divides  $F_1(m, n)P(m, n), F_2(m, n)Q(m, n), F_3(m, n)P(m, n), F_4(m, n)Q(m, n)$ . In particular,

$$\gcd(P(m, n), Q(m, n)) \mid \gcd(Am^{D+d}, Bn^{D+d})$$

Let  $A_0 = \frac{A}{\gcd(A, B)}$ ,  $B_0 = \frac{B}{\gcd(A, B)}$  then

$$\begin{aligned} \gcd(Am^{D+d}, Bn^{D+d}) &= \gcd(A, B)\gcd(A_0m^{D+d}, B_0n^{D+d}) \\ &= \gcd(A, B)\gcd(A_0, n^{D+d})\gcd(B_0, m^{D+d}) \\ &\leq \gcd(A, B)A_0B_0 = \frac{AB}{\gcd(A, B)}. \end{aligned}$$

which is exactly the  $lcm(A, B)$ . □

**Proposition 5.8.** Let  $p(x), q(x) \in \mathbb{Z}[x]$  with no common roots in  $\mathbb{C}$ . Let  $d = \max\{\deg p, \deg q\}$ . There exists a constant  $\kappa$  depending on  $p, q$  for which

$$H(\frac{p(x)}{q(x)}) \geq \kappa H(x)^d$$

**Proof.** Note that  $H(\frac{p(x)}{q(x)}) = H(\frac{P(m, n)}{Q(m, n)})$ . let  $t(m, n) = \gcd(P(m, n), Q(m, n))$ . Then

$$H(\frac{p(x)}{q(x)}) = \max\{\frac{|P(m, n)|}{t(m, n)}, \frac{|Q(m, n)|}{t(m, n)}\}$$

By the lemma,  $t(m, n) \mid lcm(A, B) = L$ . In particular,  $t(m, n) \leq L$ . Thus,

$$H(\frac{p(x)}{q(x)}) \geq \frac{1}{L} \max\{|P(m, n)|, |Q(m, n)|\}$$

Note:

$$\begin{aligned} \max\{A|m|^{D+d}, B|n|^{D+d}\} &= \max\{|F_1(m, n)P(m, n) + F_2(m, n)Q(m, n)|, |F_3(m, n)P(m, n) + F_4(m, n)Q(m, n)|\} \\ &\leq \max\{|F_1(m, n)P(m, n)| + |F_2(m, n)Q(m, n)|, |F_3(m, n)P(m, n)| + |F_4(m, n)Q(m, n)|\} \\ &\leq 2 \max\{|F_1P|, |F_2Q|, |F_3P|, |F_4Q|\} \end{aligned}$$

Note that  $F_j(m, n) = \sum_l c_l m^{D-l} n^l$

$$|F_j(m, n)| \leq \sum_l c_l m^{D-l} n^l \leq \sum_l |c_l| |m|^{D-l} |n|^l \leq \sum_l |c_l| \max\{|m|, |n|\}^D$$

Let  $C = \max_{j=1,2,3,4}\{\sum_l |c_l|\}$  then

$$|F_j(m, n)| \leq C \max(|m|, |n|)^D = CH(x)^D.$$

Thus,

$$2 \max\{|F_1P|, |F_2Q|, |F_3P|, |F_4Q|\} \leq 2CH(x)^D \max\{|P(m, n)|, |Q(m, n)|\}$$

Secondly,

$$\max\{A|m|^{D+d}, B|n|^{D+d}\} \geq \min(A, B) \max\{|m|^{D+d}, |n|^{D+d}\} = \min(A, B)H(x)^{D+d}$$

Now,

$$\begin{aligned} H(\frac{p(x)}{q(x)}) &\geq \frac{1}{L} \max\{|P(m, n)|, |Q(m, n)|\} \\ &\geq \frac{1}{2LC} H(x)^{-D} \max\{A|m|^{D+d}, B|n|^{D+d}\} \\ &\geq \frac{1}{2LC} H(x)^{-D} \min(A, B)H(x)^{D+d} \\ &\geq \kappa D^d \end{aligned}$$

where  $\kappa = \frac{\min(A,B)}{2LC}$ . □

**Definition 5.9.** Given  $p \in E(\mathbb{Q})$ , define the height of  $p$  by

$$H(p) = \begin{cases} 1, & p = O \\ H(x), & p = (x, y) \end{cases}$$

**Lemma 5.10.** Let  $P \in E(\mathbb{Q})$ ,  $P \neq O$ ,  $P = (x, y)$ . Then there exist integers  $m, n, e$  with  $x = \frac{m}{e^2}$  and  $y = \frac{n}{e^3}$  and  $\gcd(m, e) = \gcd(n, e) = 1$ .

**Proof.** Let  $P$  divide the denominator of  $x$  (or  $y$ ). Recall that there exists  $v > 0$  for which  $\text{ord}_P(x) = -2v$ ,  $\text{ord}_P(y) = -3v$ . So the denominator of  $x$  must be square, say  $x = \frac{m}{e^2}$  and then  $y$  has denominator  $e^3$ . □

How do  $m, n, e$  relate to  $H(P)$ ? We know that  $H(P) = H(x) = \max\{|m|, e^2\}$ . Thus,  $|m| \leq H(P)$  and  $|e| \leq H(P)^{\frac{1}{2}}$ . What about  $n$ ?

**Lemma 5.11.** Take  $P = (\frac{m}{e^2}, \frac{n}{e^3})$  then  $n$  is at most the height of  $H(P)^{\frac{3}{2}}$ .

**Proof.** We know that  $|n|^2 \leq H(y^2) = H(x^3 + ax^2 + bx + c)$ . Think of  $x^3 + ax^2 + bx + c$  as a rational map of degree 3 so we have  $H(x^3 + ax^2 + bx + c) \leq \kappa H(x)^3 = \kappa H(P)^3$  where  $\kappa$  is a constant depends on  $a, b, c$ . Therefore we must have

$$|n| \leq \sqrt{\kappa} H(P)^{\frac{3}{2}}$$
□

**Proposition 5.12.** Fix some  $P_0 \in E(\mathbb{Q})$ . There exists some  $\kappa$  depends only on  $E$  and  $P_0$  for which

$$H(P + P_0) \leq \kappa H(P)^2.$$

for all  $P \in E(\mathbb{Q})$ .

**Proof.** Ignore  $P_0 = O$  and  $P = O, \pm P_0$ . Let  $P_0 = (x_0, y_0)$  and  $P = (x, y)$  and  $P + P_0 = (x', y')$ . Then

$$\begin{aligned} x' &= \left( \frac{y - y_0}{x - x_0} \right)^2 - a - x_0 - x \\ &= \frac{(y - y_0)^2 - (a + x_0 + x)(x - x_0)^2}{(x - x_0)^2} \end{aligned}$$

Note that the numerator has  $y^2$  and  $-x^3$ . But note that  $y^2 - x^3 = ax^2 + bx + c$  so we substitute to get

$$x' = \frac{Ay + Bx^2 + Cx + D}{Ex^2 + Fx + G}$$

where the capital letters are some constants depending only on  $P_0$  and  $E$ . Using the result that  $x = \frac{m}{e^2}$ ,  $y = \frac{n}{e^3}$  we get

$$x' = \frac{Ane + Bm^2 + Cme^2 + De^4}{Em^2 + Fme^2 + Ge^4}$$

$$\implies H(x') \leq \max\{|Ane + Bm^2 + Cme^2 + De^4|, |Em^2 + Fme^2 + Ge^4|\}$$

Let  $\kappa_1 = \max\{|A|, \dots, |G|\}$  we have

$$\begin{aligned} H(x') &\leq \kappa_1 \max\{|ne| + |m^2| + |me^2| + |e^4|, |m^2| + |me^2| + |e^4|\} \\ &\leq \kappa_1 \max\{\kappa H(P)^{\frac{3}{2}} H(P)^{\frac{1}{2}} + H(P)^2 + H(P)(H(P)^{\frac{1}{2}})^2, \dots\} \\ &\leq \kappa' \kappa H(P)^2 \end{aligned}$$

and we are done. □

**Proposition 5.13.** There exists some  $\kappa > 0$  for which  $H(2P) \leq \kappa H(P)^4$  for all  $P \in E(\mathbb{Q})$ .

**Proof.** Trivial if  $P = O$  or if  $2P = O$ . Otherwise,  $P = (x, y)$ ,  $y \neq 0$  and  $2P = (x', y')$ .

$$x' = \left( \frac{f'(x)}{2y} \right)^2 - a - 2x = \frac{f'(x)^2 - (a + 2x)4y^2}{4y^2} = \frac{f'(x)^2 - 4(a + 2x)f(x)}{4f(x)}$$

where  $f(x) = x^3 + ax^2 + bx + c$ . To apply lower bounds on heights, need numerator and denominator to have no common roots in  $\mathbb{C}$ . If  $\alpha$  a common root, we would have  $f(\alpha) = f'(\alpha) = 0$  which is impossible on elliptic curve. We know that

$$H(x') \leq \kappa H(x)^d$$

where  $d = \max\{\deg \text{ numerator}, \deg \text{ denominator}\} = \max\{3, 4\} = 4$ . □

**Remark.** Heights grow rapidly!

Now we have all ingredients to prove Weak M-W implies M-W.

**Definition 5.14.** Let  $h(x) = \log H(x)$ ,  $x \in \mathbb{Q}$ .  $h(P) = \log H(P)$ ,  $P \in E(\mathbb{Q})$  known as the "logarithmic height".

**Theorem 5.15.** Let  $E$  be an elliptic curve over  $\mathbb{Q}$ . If  $E(\mathbb{Q})/2E(\mathbb{Q})$  is finite, then  $E(\mathbb{Q})$  is finitely generated.

**Proof.** Let  $Q_1, \dots, Q_r$  be cosets representatives for  $E(\mathbb{Q})/2E(\mathbb{Q})$ . Define 2 constants  $C_1, C_2 > 0$  so that

1.  $h(P - Q_i) \leq 2h(P) + C_1$  for all  $P \in E(\mathbb{Q})$  and all  $Q_i$ .
2.  $h(2P) \geq 4h(P) - C_2$  for all  $P \in E(\mathbb{Q})$ .

Let  $P_1, \dots, P_s$  be the set of points in  $E(\mathbb{Q})$  with  $h(P) \leq C_1 + C_2$ . Claim that  $\{P_1, \dots, P_s, Q_1, \dots, Q_r\}$  generators  $E(\mathbb{Q})$ . Use induction with base case:  $P \in \mathbb{Q}$  with  $h(P) \leq C_1 + C_2$  can be generated by  $\{P_1, \dots, P_s, Q_1, \dots, Q_r\}$ . Assume that any  $P'$  with  $h(P') < h(P)$  can be generated.

Let's prove that  $P$  lies in the subgroup generated by  $\{P_1, \dots, P_s, Q_1, \dots, Q_r\}$ . There is some  $Q_i$  so that  $P - Q_i = 2P'$ . Suppose that  $h(P) > C_1 + C_2$ . We know that

$$4h(P') - C_2 \leq h(2P') = h(P - Q_i) \leq 2h(P) + C_1$$

This means  $h(P') \leq \frac{1}{2}h(P) + \frac{C_1 + C_2}{4} < \frac{3}{4}h(P) < h(P)$ . Since  $P'$  can be generated we see that  $P = Q_i + 2P'$  can be generated so  $E(\mathbb{Q})$  is finitely generated.  $\square$

**Exercise.** Suppose that  $\frac{2a}{b}, \frac{c}{d} \in \mathbb{Q}$  lowest terms. Suppose that  $x^2 + \frac{2a}{b}x + \frac{c}{d}$  has rational roots  $r_1, r_2$ . Prove that

$$H(r_1)H(r_2) \geq \max\{H(\frac{a}{b}), H(\frac{c}{d})\}$$

**Proof.** First of all note that  $r_1 r_2 = \frac{c}{d}$  so that

$$H(\frac{c}{d}) = H(r_1 r_2) \leq H(r_1)H(r_2)$$

In addition, since  $r_1 + r_2 = -\frac{2a}{b}$  we must have

$$\begin{aligned} H(-\frac{2a}{b}) &= H(r_1 + r_2) \leq 2H(r_1)H(r_2) \\ &\implies \max\{|a|, \frac{|b|}{2}\} \leq H(r_1)H(r_2) \end{aligned}$$

Thus, it's enough to prove  $|b| \leq H(r_1)H(r_2)$ . But this is clear since  $r_1 + r_2 = -\frac{2a}{b}$  implies the denominator one of  $r_1$  or  $r_2$  is at least  $b$ .  $\square$

**Proof of Weak Mordell-Weil.** Note  $P = (x, y)$ ,  $2P = (x', y')$  and  $y^2 = x^3 + ax^2 + bx + c$  then

$$\begin{aligned} x' &= \frac{f'(x)^2 - 4(a+2x)f(x)}{4f(x)} \\ y' &= \frac{f'(x)}{2y}(x' - x) + y \end{aligned}$$

The map  $x \mapsto x'$  is a rational function of degree 4. Want to decompose the doubling map into a composition of 2 simpler rational functions. We need to find 2 rational maps  $\varphi, \psi$  such that

$$\varphi(\psi(P)) = 2P \text{ for all } P \in E(\mathbb{Q})$$

Suppose that  $E$  corresponds to  $\mathbb{C}/\Lambda$ ,  $\Lambda$  with basis  $\{\omega_1, \omega_2\}$ . In  $\mathbb{C}/\Lambda$ , doubling map is very simple:  $z \pmod{\Lambda} \mapsto 2z \pmod{\Lambda}$ . Let  $\bar{\Lambda}$  be the lattice with basis  $\{\frac{\omega_1}{2}, \omega_2\}$ ,  $\bar{\Lambda}$  be the lattice with basis  $\{\frac{\omega_1}{2}, \frac{\omega_2}{2}\}$ . The first map is  $\varphi$ , the second map is  $\bar{\varphi}$ , the third map is  $\gamma$  then define  $\psi = \gamma \circ \bar{\varphi}$ . Good way to see "bigger picture", bad if we want to use  $x, y$  coordinates. More convenient to work with  $E$  directly. Need to determine  $\varphi, \psi, E(\Lambda)$ . Note that

$$\wp(z; \bar{\Lambda}) = \wp(z; \Lambda) + \wp(z + \frac{\omega_1}{2}; \Lambda)$$

Let us define all maps over  $\mathbb{Q}$ , if  $y^2 = x^3 + ax^2 + bx + c$  has a rational root  $x_0$  then  $(x_0, 0)$  a point of order 2. Shift variables to assume that the root is  $x_0 = 0$ . Take  $E$  of the form

$$y^2 = x^3 + ax^2 + bx$$

Defining  $\bar{E}$  and maps  $\varphi : E \rightarrow \bar{E}$ ,  $\psi : \bar{E} \rightarrow E$ . Define  $\bar{E}$  by Weierstrass equations

$$\bar{E} : y^2 = x^3 + \bar{a}x^2 + \bar{b}x$$

where  $\bar{a} = -2a$  and  $\bar{b} = a^2 - 4b$ . We claim that  $\bar{E}$  is an elliptic curve, the discriminant

$$\Delta(x^3 + Ax^2 + Bx) = B^2(A^2 - 4B)$$

Thus,

$$\begin{aligned} \Delta(x^3 + \bar{a}x^2 + \bar{b}x) &= \bar{b}^2(\bar{a}^2 - 4\bar{b}) \\ &= (a^2 - 4b)^2((-2a)^2 - 4(a^2 - 4b)) \\ &= (a^2 - 4b)^2 \cdot 16b \end{aligned}$$

Since  $E$  was elliptic, we know that  $b^2(a^2 - 4b) \neq 0$ . Thus,  $\Delta(\bar{E}) \neq 0$ . What happens if we repeat this? Define  $\bar{\bar{E}} : y^2 = x^3 + \bar{\bar{a}}x^2 + \bar{\bar{b}}x$  where  $\bar{\bar{a}} = -2\bar{a} = 4a$  and  $\bar{\bar{b}} = \bar{a}^2 - 4\bar{b}^2 = (-2a)^2 - 4(a^2 - 4b) = 16b$ . Therefore,

$$\bar{\bar{E}} : y^2 = x^3 + 4ax^2 + 16bx$$

There is a map  $\bar{\bar{E}} \rightarrow E$ ,  $(x, y) \mapsto (4x, 8y)$ .  $\bar{\bar{E}}$  and  $E$  are isomorphic as elliptic curves. Defining  $\varphi : E \rightarrow \bar{E}$ , let  $T = (0, 0)$ . Defining  $\varphi(O) = O$ ,  $\varphi(T) = O$ . For other  $P = (x, y)$ ,  $\varphi(P) = (\bar{x}, \bar{y})$ ,  $\bar{x} = x + a + \frac{b}{x} = \frac{y^2}{x^2}$  and  $\bar{y} = y \frac{x^2 - b}{x}$ . Claim:  $\varphi$  maps  $E$  to  $\bar{E}$ . Suppose  $(x, y) \in E$ ,

$$\begin{aligned} \bar{x}^3 + \bar{a}\bar{x}^2 + \bar{b}\bar{x} &= \frac{y^6}{x^6} - 2a\frac{y^4}{x^4} + (a^2 - 4b)\frac{y^2}{x^2} \\ &= \frac{y^2}{x^6}(y^4 - 2ay^2x^2 + (a^2 - 4b)x^4) = \frac{y^2}{x^6}((y^2 - ax^2)^2 - 4bx^4) = \frac{y^2}{x^6}((x^3 + bx)^2 - 4bx^4) \\ &= \frac{y^2}{x^4}((x^2 + b)^2 - 4bx^2) = \frac{y^2(x^2 - b)^2}{x^4} \end{aligned}$$

Defining a map  $\psi : \bar{E} \rightarrow E$  to get a map

$$\bar{\varphi} : \bar{E} \rightarrow \bar{\bar{E}}$$

by  $\psi(\bar{O}) = O$ ,  $\psi(\bar{T}) = O$  where  $\bar{T} = (0, 0)$ . And

$$\psi(\bar{x}, \bar{y}) = \left( \frac{\bar{y}^2}{4\bar{x}^2}, \bar{y} \cdot \frac{\bar{x}^2 - \bar{b}}{\bar{x}^2} \right)$$

We need the following proposition. Ultimately, we want to understand the image  $2E(\mathbb{Q})$  of the composition  $\psi \circ \varphi$ . Towards that, study  $\text{Im } \varphi$ ,  $\text{Im } \phi$ . Let  $\Gamma := E(\mathbb{Q})$ ,  $\bar{\Gamma} := \bar{E}(\mathbb{Q})$ . We want to show that  $\Gamma/2\Gamma$  is finite. Idea:  $\bar{\Gamma}/\varphi(\Gamma)$ ,  $\Gamma/\psi(\bar{\Gamma})$  are finite.  $\square$

**Proposition 5.16.** We have the following properties:

1.  $\varphi, \psi$  homomorphisms.
2.  $\ker\{O, T\}$  and  $\ker\{\bar{O}, \bar{T}\}$  by definition.
3.  $\psi(\varphi(P)) = 2P$  for all  $P \in E(\mathbb{Q})$ ,  $\varphi(\psi(\bar{P})) = 2\bar{P}$  for all  $\bar{P} \in \bar{E}(\mathbb{Q})$ .

**Proposition 5.17.** Let  $\bar{P} \in \bar{\Gamma}$ , if  $\bar{P} = \bar{O}$  then  $\bar{P} \in \varphi(\Gamma)$ .  $\bar{P} = \bar{T} = (0, 0) \in \varphi(\Gamma)$  if and only if  $\bar{b} = a^2 - 4b$  is a perfect square.  $\bar{P} = (\bar{x}, \bar{y}), \bar{x} \neq 0$  then  $\bar{P} \in \varphi(\Gamma)$  if and only if  $\bar{x}$  is the square of a rational i.e.  $x \in (\mathbb{Q}^\times)^2$ .

**Proof.**  $\bar{O} \in \varphi(P)$  is trivial since  $\varphi(O) = \bar{O}$ .  $\bar{T} \in \varphi(\Gamma)$  if and only if  $\bar{T} = \varphi(x, y)$  for some  $(x, y) \in \Gamma$  if and only if  $(0, 0) = \varphi(x, y) = (\frac{y^2}{x^2}, y \frac{x^2 - b}{x^2})$  i.e. there exists some  $(x, y) \in P$  such that  $x \neq 0$  and  $y = 0$ . Thus,  $x^3 + ax^2 + bx$  has a non-zero root in  $\mathbb{Q}$ .  $x^2 + ax + b$  has a rational root. Thus

$$\frac{-a \pm \sqrt{a^2 - 4b}}{2a} \in \mathbb{Q} \implies a^2 - 4b \in (\mathbb{Q}^\times)^2.$$

Fix some  $\bar{P} = (\bar{x}, \bar{y})$ . If  $\bar{P} \in \varphi(\Gamma)$ , then  $\bar{x} = \frac{y^2}{x^2}$  for some  $(x, y) \in \Gamma$  which implies  $x \in (\mathbb{Q}^\times)^2$ . Conversely, suppose that  $\bar{x} = \omega^2$ ,  $\omega \neq 0$ ,  $\omega \in \mathbb{Q}$ . We want some  $(x, y)$  with  $\varphi((x, y)) = (\bar{x}, \bar{y})$ . Actually, we expect 2 preimages, since  $\ker \varphi = \{O, T\}$ . These points are  $P_1 = (x_1, y_1)$ ,  $P_2 = (x_2, y_2)$  where

$$\begin{aligned} x_1 &= \frac{1}{2}(\omega^2 - a + \frac{\bar{y}}{\omega}), & y_1 &= x_1\omega. \\ x_2 &= \frac{1}{2}(\omega^2 - a - \frac{\bar{y}}{\omega}), & y_2 &= -x_2\omega. \end{aligned}$$

Want to show that  $\varphi(P_1) = \varphi(P_2) = \bar{P}$  but it's best to try to play  $P_1, P_2$  off each other and treat at the same time. First note that

$$\begin{aligned} x_1x_2 &= \frac{1}{4}(\omega^2 - a + \frac{\bar{y}}{\omega})(\omega^2 - a - \frac{\bar{y}}{\omega}) = \frac{1}{4}((\omega^2 - a)^2 - \frac{\bar{y}^2}{\omega^2}) \\ &= \frac{1}{4}((\bar{x} - a)^2 - \frac{\bar{y}^2}{\bar{x}}) = \frac{1}{4}(\frac{\bar{x}^3 - 2a\bar{x}^2 + a^2\bar{x} - \bar{y}^2}{\bar{x}}) \end{aligned}$$

Remember  $(\bar{x}, \bar{y}) \in \bar{\Gamma}$ . So  $\bar{y}^2 = \bar{x}^3 + \bar{a}\bar{x}^2 + \bar{b}\bar{x}$ . Thus,

$$x_1x_2 = \frac{1}{4}(\frac{a^2\bar{x} - \bar{b}\bar{x}}{\bar{x}}) = b$$

since  $\bar{b} = a^2 - 4b$ . To complete the proof, enough to show

1.  $P_1, P_2 \in \Gamma$
2.  $\varphi(P_1) = \varphi(P_2) = \bar{P}$ .

For 1, want to prove  $y_i^2 = x_i^3 + ax_i^2 + bx_i$  for  $i = 1, 2$ .

$$\frac{y_i^2}{x_i^2} = x_i + a + \frac{b}{x_i}$$

LHS is always  $\omega^2$ . In addition, since  $x_1x_2 = b$  we have  $\omega^2 = x_1 + a + x_2$ . But this is now obviously by definition of  $x_1, x_2$ . For 2,  $\varphi(P_i) = \bar{P}$  holds if and only if  $\frac{y_i^2}{x_i^2} = \bar{x}$  and  $y_i \frac{x_i^2 - b}{x_i^2} = \bar{y}$ . Condition on  $\bar{x}$  follows immediately, as LHS is

$\omega^2$ . For condition on  $\bar{y}$ :

$$\begin{aligned} y_1 \frac{x_1^2 - b}{x_1^2} &= y_1 \frac{x_1^2 - x_1 x_2}{x_1^2} = x_1 \omega \frac{x_1^2 - x_1 x_2}{x_1^2} = \omega(x_1 - x_2) \\ y_2 \frac{x_2^2 - b}{x_2^2} &= y_2 \frac{x_2^2 - x_1 x_2}{x_2^2} = (-x_2 \omega) \frac{x_2^2 - x_1 x_2}{x_2^2} = \omega(x_1 - x_2). \end{aligned}$$

But  $x_1 - x_2 = \frac{\bar{y}}{\omega}$  by definition then  $\omega(x_1 - x_2) = \bar{y}$ . □

**Definition 5.18.** Let

$$\alpha(P) = \begin{cases} 1, & P = O \\ b, & P = T = (0, 0) \\ x, & P = (x, y), x \neq 0 \end{cases} \pmod{(\mathbb{Q}^\times)^2}$$

and  $\alpha : \Gamma \rightarrow \mathbb{Q}^\times / (\mathbb{Q}^\times)^2$ .

**Proposition 5.19.**

1.  $\alpha : \Gamma \rightarrow \mathbb{Q}^\times / (\mathbb{Q}^\times)^2$  is a homomorphism.
2.  $\ker \alpha = \psi(\bar{\Gamma})$ .
3. If  $p_1, \dots, p_s$  are prime divisors of  $b$  then  $\text{Im } \alpha \subset \alpha(\Gamma)$  is contained in  $\{\pm p_1^{\varepsilon_1} \cdots p_s^{\varepsilon_s} : \varepsilon_j = 0, 1\} \subseteq \mathbb{Q}^\times / (\mathbb{Q}^\times)^2$

**Proof.**

1. It suffices to show that if  $P_1 + P_2 + P_3 = O$  then  $\alpha(P_1)\alpha(P_2)\alpha(P_3) \equiv 1 \pmod{(\mathbb{Q}^\times)^2}$ . Suppose that  $P_1, P_2, P_3$  lie on a line,  $y = \lambda x + \nu$  (ignoring some edge cases). Let  $P_j = (x_j, y_j)$ , the  $x_j$  are roots of  $(\lambda x + \nu)^2 = ax^3 + ax^2 + bx$ . This implies  $x_1 x_2 x_3 = 0$ . Then  $\alpha(P_1)\alpha(P_2)\alpha(P_3) = x_1 x_2 x_3 = v^2 \equiv 1 \pmod{(\mathbb{Q}^\times)^2}$ .
2. Follows immediately from the fact that  $\psi(\bar{\Gamma}) = \{P \in \Gamma : x \in (\mathbb{Q}^\times)^2\}$ .
3. Recall that any  $P = (x, y) \in \Gamma$  has  $(x, y) = (\frac{m}{e^2}, \frac{n}{e^3})$  for some  $m, n, e$  integers.  $(x, y) \in \Gamma$  implies

$$\begin{aligned} \left(\frac{n}{e^3}\right)^2 &= \left(\frac{m}{e^2}\right)^3 + a\left(\frac{m}{e^2}\right)^2 + b\left(\frac{m}{e^2}\right) \\ \implies n^2 &= m^3 + am^2 e^2 + bme^4 = m(m^2 + ame^2 + be^4) \end{aligned}$$

Let  $p \mid m$  be prime and suppose that  $p \nmid b$  then  $p \nmid m^2 + ame^2 + be^4$ . We know that  $\text{ord}_p(n^2) = \text{ord}_p(m) + \text{ord}_p(m^2 + ame^2 + be^4)$ . Thus,  $p \nmid b$  implies  $\text{ord}_p(m)$  is even. Only prime divisors of  $m$  with odd multiplicity are primes divide  $b$ .  $m = \pm m_0^2 \cdot p_1^{\varepsilon_1} \cdots p_s^{\varepsilon_s}$  where  $\varepsilon_i \in \{0, 1\}$  and  $p_1, \dots, p_s$  are primes dividing  $b$ . Then

$$\alpha(P) = x = \frac{m}{e^2} \equiv m \equiv \pm m_0^2 \cdot p_1^{\varepsilon_1} \cdots p_s^{\varepsilon_s} \equiv \pm p_1^{\varepsilon_1} \cdots p_s^{\varepsilon_s} \pmod{(\mathbb{Q}^\times)^2}$$

and we are done. □

**Corollary 5.20.**  $\Gamma / \ker \alpha \cong \text{Im } \alpha$  which means

$$\Gamma / \psi(\bar{\Gamma}) \hookrightarrow_{\text{inj}} \{\pm p_1^{\varepsilon_1} \cdots p_s^{\varepsilon_s}\} \subseteq \mathbb{Q}^\times / (\mathbb{Q}^\times)^2$$

Thus,  $|\Gamma / \psi(\bar{\Gamma})| \leq 2^{s+1}$ , finite. So the index  $[\Gamma : \psi(\bar{\Gamma})] < \infty$ .

Likewise, can do all of this to define  $\bar{\alpha} : \bar{\alpha} \rightarrow \mathbb{Q}^\times / (\mathbb{Q}^\times)^2$ . And

$$\alpha(\bar{P}) = \begin{cases} 1 & \pmod{(\mathbb{Q}^\times)^2} & \bar{P} = \bar{O} \\ \bar{b} & \pmod{(\mathbb{Q}^\times)^2} & \bar{P} = \bar{T} = (0, 0) \\ \bar{x} & \pmod{(\mathbb{Q}^\times)^2} & \bar{P} = (\bar{x}, \bar{y}), \bar{x} \neq 0. \end{cases}$$

As what we have done previously, we would get  $[\bar{\Gamma} : \varphi(\Gamma)] < \infty$ . To finish, we claim that  $[\Gamma : \psi(\bar{\Gamma})] < \infty$  and  $[\bar{\Gamma} : \varphi(\Gamma)] < \infty$  imply  $[\Gamma : 2\Gamma] = [\Gamma : \psi(\varphi(\Gamma))] < \infty$ .

**Lemma 5.21.** Let  $A, B$  be abelian groups. Suppose  $\varphi : A \rightarrow B$  and  $\psi : B \rightarrow A$  are homomorphisms, with  $\psi(\varphi(a)) = 2a$ ,  $\varphi(\psi(b)) = 2b$  for all  $a \in A$ ,  $b \in B$ . If  $[A : \psi(B)] < \infty$  and  $[B : \varphi(A)] < \infty$ , then  $[A : 2A]$  is finite, in particular

$$[A : 2A] < [A : \psi(B)] \cdot [B : \varphi(A)]$$

**Proof.** Know that  $A/\psi(B)$  is finite. Let  $a_1, \dots, a_n$  be coset representatives. Let  $b_1, \dots, b_m$  be coset representatives for  $B/\varphi(A)$ . We claim that  $S := \{a_i + \psi(b_j)\}$  represents all cosets in  $A/2A$ . Let  $a \in A$ . Can write  $a + \psi(B) = a_i + \psi(B)$  as cosets. Some  $a_i, a - a_i \in \psi(B)$  implies  $a - a_i = \psi(b)$  for some  $b \in B$ . Can write  $b + \varphi(A) = b_j + \varphi(A)$  as cosets. Then we have  $b - b_j = \varphi(a')$  for some  $a' \in A$ . Then

$$\begin{aligned} a &= a_i + \psi(b) = a_i + \psi(b_j + \varphi(a')) = a_i + \psi(b_j) + \psi(\varphi(a')) = a_i + \psi(b_j) + 2a' \\ \implies a + 2A &= a_i + \psi(b_j) + 2A \end{aligned}$$

Then  $a$  is represented by coset representatives of  $a_i + \psi(b_j) \in S$ . □

# 6 Computing Ranks of Elliptic Curves

Structure for finitely generated abelian groups are

$$E(\mathbb{Q}) \cong \underbrace{\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}}_r \oplus \underbrace{\mathbb{Z}/p_1^{e_1}\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/p_k^{e_k}\mathbb{Z}}_{E(\mathbb{Q})_{\text{tors}}}$$

then  $E(\mathbb{Q}) \cong E(\mathbb{Q})_{\text{tors}} \oplus \mathbb{Z}^r$ .

**Definition 6.1.**  $r$  is called the rank of  $E$ .

No known algorithm which can determine the rank of a general elliptic curve. But we have tools in certain cases, like when  $y^2 = x^3 + ax^2 + bx$ ,  $a, b, c \in \mathbb{Z}$  (curve has rational point of order 2). Recall  $\Gamma = E(\mathbb{Q})$ ,

$$\frac{\Gamma}{2\Gamma} \cong \frac{\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}}{2\mathbb{Z} \oplus \cdots \oplus 2\mathbb{Z}} \oplus \frac{C_{p_1^{e_1}}}{2C_{p_1^{e_1}}} \oplus \cdots \oplus \frac{C_{p_k^{e_k}}}{2C_{p_k^{e_k}}} \cong (\mathbb{Z}/2\mathbb{Z})^r \oplus \frac{C_{p_1^{e_1}}}{2C_{p_1^{e_1}}} \oplus \cdots \oplus \frac{C_{p_k^{e_k}}}{2C_{p_k^{e_k}}}$$

If  $p_j$  is odd, then  $x \mapsto 2x \pmod{p_j^{e_j}}$  is an isomorphism so

$$\frac{C_{p_j^{e_j}}}{2C_{p_j^{e_j}}} \cong 1$$

For  $p = 2$ , kernel of  $x \mapsto 2x$  is  $\{0, \frac{2^e}{2}\}$ . Then

$$\Gamma/2\Gamma \cong (\mathbb{Z}/2\mathbb{Z})^{r+\#\{j \in [1, n] : p_j=2\}} = 2^r E(\mathbb{Q})[2]$$

$$[\Gamma : 2\Gamma] = 2^r \cdot |\Gamma[2]|.$$

Note that

$$|\Gamma[2]| = \begin{cases} 2, & a^2 - 4b \notin (\mathbb{Q}^\times)^2 \\ 4, & a^2 - 4b \in (\mathbb{Q}^\times)^2 \end{cases}$$

Computing  $[\Gamma : 2\Gamma]$ : we have chain inclusions

$$2\Gamma \subseteq \psi(\bar{\gamma}) \subseteq \Gamma$$

then

$$\frac{\Gamma/2\Gamma}{\psi(\bar{\Gamma}/2\Gamma)} \cong \Gamma/\psi(\bar{\Gamma})$$

Hence,

$$[\Gamma : 2\Gamma] = [\psi(\bar{\Gamma}) : 2\Gamma][\Gamma : \psi(\bar{\Gamma})] = [\psi(\bar{\Gamma}) : \psi(\varphi(\Gamma))][\Gamma : \psi(\bar{\Gamma})]$$

Let's reduce  $[\psi(\bar{\Gamma}) : \psi(\varphi(\Gamma))]$  to  $[\bar{\Gamma} : \varphi(\Gamma)]$ .

**Lemma 6.2.** Let  $\psi : A \rightarrow A'$  be a homomorphisms of abelian groups. Let  $B \subseteq A$  be a subgroup, then

1.  $\psi(B) \subseteq A'$  is a subgroup.
2. If all indices are finite, then

$$[\psi(A) : \psi(B)] = \frac{[A : B]}{[\ker \psi : B \cap \ker \psi]}$$

**Proof.** We know that  $\psi : A \rightarrow A'$  and  $(\text{mod } \psi(B)) : A' \rightarrow A'/\psi(B)$  given by

$$a \mapsto \psi(a) \mapsto \psi(a) + \psi(B)$$

kernel of the composition contains  $B$ . Define a quotient homomorphism

$$\bar{\psi} : A/B \rightarrow \psi(A)/\psi(B)$$

$$a + B \mapsto \psi(a) + \psi(B)$$

$\bar{\psi}$  is clearly surjective so

$$(A/B)/\ker \bar{\psi} \cong \psi(A)/\psi(B)$$

As for  $\ker \bar{\psi}$ ,

$$\begin{aligned} a + B \in \ker \bar{\psi} &\Leftrightarrow \psi(a) = \psi(B) \\ &\Leftrightarrow \psi(a) = \psi(b), b \in B \\ &\Leftrightarrow \psi(a - b) = 0 \\ &\Leftrightarrow a - b \in \ker \psi \\ &\Leftrightarrow a \in \ker \psi \oplus B \end{aligned}$$

Thus,

$$\ker \bar{\psi} = \frac{\ker \psi \oplus B}{B} \cong \ker \psi / (\ker \psi \cap B).$$

Then  $|\ker \bar{\psi}| = [\ker \psi : \ker \psi \cap B]$ . □

Then if  $r$  is the rank of  $E$ ,

$$\begin{aligned} 2^r &= \frac{[\Gamma : 2\Gamma]}{|\Gamma[2]|} = \frac{[\psi(\bar{\Gamma}) : \psi(\varphi(x))][\Gamma : \psi(\bar{\Gamma})]}{|\Gamma[2]|} \\ &= \frac{[\Gamma : \psi(\bar{\Gamma})]}{|\Gamma[2]|} \cdot \frac{[\bar{\Gamma} : \varphi(\Gamma)]}{|\ker \psi : \varphi(\Gamma) \cap \ker \psi|} \end{aligned}$$

We have an explicit implication of  $\psi$ :

$$\ker \psi = \{O, T = (0, 0)\}$$

Thus,

$$[\ker \psi : \varphi(P) \cap \ker \psi] = \begin{cases} 1, & a^2 - 4b \in (\mathbb{Q}^\times)^2 \\ 2, & \text{otherwise} \end{cases}$$

Recall

$$\begin{aligned} |\Gamma[2]| &= \begin{cases} 2, & a^2 - 4b \notin (\mathbb{Q}^\times)^2 \\ 4, & a^2 - 4b \in (\mathbb{Q}^\times)^2 \end{cases} \\ \implies 2^r &= \frac{[\Gamma : \psi(\bar{\Gamma})] \cdot [\bar{\Gamma} : \varphi(\Gamma)]}{4} \end{aligned}$$

We can write the indices here in terms of  $\alpha, \bar{\alpha}$  where

$$\alpha : \Gamma \rightarrow \mathbb{Q}^\times / (\mathbb{Q}^\times)^2, \ker \alpha = \psi(\bar{\Gamma})$$

$$\bar{\alpha} : \bar{\Gamma} \rightarrow \mathbb{Q}^\times / (\mathbb{Q}^\times)^2, \ker \bar{\alpha} = \varphi(\Gamma)$$

Then  $[\Gamma : \psi(\bar{\Gamma})] = |\text{Im } \alpha|$ ,  $[\bar{\Gamma} : \varphi(\Gamma)] = |\text{Im } \bar{\alpha}|$ . Our final rank formula is the

$$2^r = \frac{|\text{Im } \alpha| |\text{Im } \bar{\alpha}|}{4}$$

Recall

$$\text{Im } \alpha = \{\pm p_1^{\varepsilon_1} \cdots p_s^{\varepsilon_s} \text{ where } p_1, \dots, p_s \text{ are prime divisors of } b\}$$

and  $\varepsilon_j \in \{0, 1\}$ . And  $\text{Im } \alpha$  contains  $1, b$ . Otherwise,  $\alpha(P) = x$ ,  $P = (x, y)$  so we want to know which rationals can appear as  $x$ -coordinates of rational points up to squares. Let  $P = (\frac{m}{e^2}, \frac{n}{e^3})$ ,  $m, n, e \in \mathbb{Z}$ ,  $(m, e) = (n, e) = 1$ . By the equation of  $E$ :

$$\begin{aligned} \left(\frac{n}{e^3}\right)^2 &= \left(\frac{m}{e^2}\right)^3 + a\left(\frac{m}{e^2}\right)^2 + b\left(\frac{m}{e^2}\right). \\ n^2 &= m^3 + am^2e^2 + bme^4 = m(m^2 + ame^2 + be^4) \end{aligned}$$

Let  $b_1 = \gcd(m, m^3 + ame^2 + be^4) = \gcd(m, be^4) = \gcd(m, n)$ . Then  $m = b_1 m_1$ ,  $b = b_1 b_2$ .

$$\implies n^2 = b_1^2 m_1 (b_1 m_1^2 + am_1 e^2 + b_2 e^4)$$

Then  $b_1^2 \mid n^2$  so  $b_1 \mid n$ . Write  $n = b_1 n_1$  we have

$$n_1^2 = m_1 (b_1 m_1^2 + am_1 e^2 + b_2 e^4)$$

Thus,  $m_1$  and  $b_1 m_1^2 + am_1 e^2 + b_2 e^4$  are both integer squares. Let  $m_1 = M_1^2$  and  $b_1 m_1^2 + am_1 e^2 + b_2 e^4 = N^2$ . In terms of our new variables

$$P = \left(\frac{b_1 M^2}{e^2}, \frac{b_1 M N}{e^3}\right)$$

This means  $\alpha(P) = \frac{b_1 M^2}{e^2} \equiv b_1 \pmod{(\mathbb{Q}^\times)^2}$  so  $b_1 \in \text{Im } \alpha$ . In summary:

**Proposition 6.3.** Let  $P = (x, y) \in \Gamma$ ,  $x \neq 0$ . Then there exists a factorization  $b = b_1 b_2$  and integers  $N, M, e$ ,  $M \neq 0$  so that

$$\begin{aligned} N^2 &= b_1 M^4 + a M^2 e^2 + b_2 e^4 \\ \gcd(b_1, e) &= \gcd(M, e) = \gcd(N, e) = 1 \\ \gcd(M, b_2) &= \gcd(M, N) = 1. \end{aligned}$$

Conversely, if  $(N, M, e)$  satisfies the above, then

$$P = \left(\frac{b_1 M^2}{e^2}, \frac{b_1 M N}{e^3}\right) \in \Gamma, \alpha(P) = b_1$$

Here is a procedure for computing the rank:

1. For each factorization  $b = b_1 b_2$ , attempt to find  $(N, M, e)$  as in proposition. If we succeed, then  $b_1 \in \text{Im } \alpha$ . If we fail for  $b_1$  and all other divisions  $b'_1 \mid b$  with  $b_1 \equiv b'_1 \pmod{(\mathbb{Q}^\times)^2}$ , then  $b_1 \notin \text{Im } \alpha$ .
2. Repeat for  $\text{Im } \bar{\alpha}$ .
3. Use the formulas

$$2^r = \frac{|\text{Im } \alpha| |\text{Im } \bar{\alpha}|}{4}$$

**Example.** Consider  $y^2 = x^3 - x$  then  $b = -1$ . Start with  $\text{Im } \alpha$ , we know that  $\text{Im } \alpha = \{\pm 1\}$ , and  $1, b \in \text{Im } \alpha$ , so  $\text{Im } \alpha = \{\pm 1\}$ . Next, consider  $\bar{\alpha}$ . Here,  $\bar{E} : y^2 = x^3 + 4x$ ,  $\bar{b} = 0 - 4(-1) = 4$ . Then  $\text{Im } \bar{\alpha} = \{\pm 1, \pm 2\}$ . Consider

whether  $-1 \in \text{Im } \bar{\alpha}$  if we can solve

$$N^2 = -M^4 - 4e^4, \quad N^2 = -4M^4 - e^4$$

but neither of the equations are solvable by considering about the sign. Thus,  $-1 \notin \text{Im } \bar{\alpha}$ . Next, consider if  $-2 \in \text{Im } \bar{\alpha}$ .

$$N^2 = -2M^4 - 2e^4$$

again, no solutions. Thus,  $-2 \notin \text{Im } \bar{\alpha}$ . Is  $2 \in \text{Im } \bar{\alpha}$ ?

$$N^2 = 2M^4 + 2e^4$$

and this equation is solvable, in particular,  $(2, 1, 1)$  is a solution. Thus,  $2 \in \text{Im } \bar{\alpha}$  i.e.  $\text{Im } \bar{\alpha} = \{1, 2\}$  so

$$2^r = \frac{2 \cdot 2}{4} = 1$$

the rank of  $E$  is zero which means  $E(\mathbb{Q}) = E(\mathbb{Q})_{\text{tors}}$ . Therefore,  $E(\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ .

**Proposition 6.4.** Let  $b_1$  be a square-free divisor of  $b$ ,  $b_1 \neq 1, b$ . Then  $b_1 \in \text{Im } \alpha$  if and only if there exist  $N, M, e \in \mathbb{Z}$  with  $M \neq 0$  such that

$$N^2 = b_1 M^4 + a M^2 e^2 + (b/b_1) e^4$$

and  $\gcd(MNb, e) = 1$ .

**Proof.** ( $\Leftarrow$ ) : Check that  $P = (\frac{b_1 M^2}{e^2}, \frac{b_1 MN}{e^2}) \in \Gamma$ . Clearly  $\alpha(P) = \frac{b_1 M^2}{e^2} \equiv b \pmod{(\mathbb{Q}^\times)^2}$ .

( $\Rightarrow$ ) : Suppose  $b_1 \in \text{Im } \alpha$ . There exists  $P$  such that  $\alpha(P) = b \pmod{(\mathbb{Q}^\times)^2}$ . By earlier proposition, there exists  $b'_1/b, b'_1 \equiv b \pmod{(\mathbb{Q}^\times)^2}$  and such that there exists  $(N_0, M_0, e_0)$ ,  $M_0 \neq 0$  such that... Since  $b_1$  is square free,  $b'_1, b \in \mathbb{Z}$  and  $b_1^2 = s^2 b_1, s \in \mathbb{Q}^\times$ . In fact,  $s \in \mathbb{Z}$

$$\begin{aligned} N_0^2 &= b_1^2 M_0^4 + a M_0^2 e_0^2 + \left(\frac{b}{b'_1}\right) e_0^4 \\ N_0^2 &= b_1 s^2 M_0^4 + a M_0^2 e_0^2 + \left(\frac{b}{s^2 b_1}\right) e_0^4 \\ (s N_0)^2 &= b_1 (s M_0)^4 + a (s M_0)^2 e_0^2 + \left(\frac{b}{b_1}\right) e_0^4 \end{aligned}$$

Take  $(N, M, e) = (s N_0, s M_0, e_0)$ , check that  $\gcd(MNb, e) = \gcd(s^2 M_0 N_0 b_1, e_0)$ . Have  $e_0$  coprime to  $M_0 N_0 b'_1$   $\square$

**Example.** Consider  $E : y^2 = x^3 - 5x$ . Compute the rank of  $E$ :

1. Compute  $\text{Im } \alpha$ : since  $b = -5$  we have

$$\text{Im } \alpha \subseteq \{\pm 1, \pm 5\}$$

We know that  $1, -5 \in \text{Im } \alpha$ . Do  $-1, 5 \in \text{Im } \alpha$ :

$$-1 : N^2 = -M^4 + 5e^4$$

Since  $\text{Im } \alpha$  is a subgroup of  $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ , it's closed under multiplication.

$$-1, -5 \in \text{Im } \alpha \implies 5 = (-1)(-5) \in \text{Im } \alpha$$

2. Compute  $\text{Im } \bar{\alpha}$ ,

$$\begin{aligned} \bar{E} : y^2 &= x^3 + 20x \\ \text{Im } \bar{\alpha} &\subseteq \{\pm 1, \pm 2, \pm 5, \pm 10\} \\ -1 : N^2 &= -M^4 - 20e^4 \end{aligned}$$

has no solutions by signs. Similarly, if  $b_1 < 0$  then  $b_1 \notin \text{Im } \bar{\alpha}$  by signs. Thus,

$$2 : N^2 = 2M^4 + 10e^4$$

Reduce mod 5 we have

$$N^2 \equiv 2M^4 \pmod{5}$$

Since 2 is not a quadratic residue mod 5, this implies  $N, M \equiv 0 \pmod{5}$ . Then  $25 \mid N^2, 25 \mid 2M^4, 25 \mid 10e^4$  i.e.  $5 \mid e$  contradicting  $\gcd(MNb, e) = 1$ . Then  $2 \notin \text{Im } \bar{\alpha}$ . For 10, suppose  $10 \in \text{Im } \bar{\alpha}$ . Since  $5 \in \text{Im } \bar{\alpha}, 5 \cdot 10 \in \text{Im } \bar{\alpha}, 5 \cdot 10 \equiv 2 \in \text{Im } \bar{\alpha}$  a contradiction.

3. Since  $\text{Im } \alpha = \{\pm 1, \pm 5\}$ ,  $\text{Im } \bar{\alpha} = \{1, 5\}$  we have

$$2^r = \frac{4 \cdot 2}{2} = 2 \implies r = 1$$

**Remark.**

1. Closure under multiplication. If  $b_1 \in \text{Im } \alpha$  then  $b'_1 \in \text{Im } \alpha \Leftrightarrow b_1 b'_1 \in \text{Im } \alpha$ .
2. Signs.  $N^2 = b_1 M^4 + a M^2 e^2 + (b/b_1) e^4$ . If  $a \leq 0, b > 0$ . Then  $b_1 < 0 \implies b_1 \notin \text{Im } \alpha$ .
3. Quadratic Residues. Can reduce  $N^2 \equiv b_1 M^4 + a M^2 e^2 + (b/b_1) e^4 \pmod{\text{various primes}}$ .

**Example.** Consider  $E : y^2 = x^3 + 10x^2 + 8x$ . Note that  $\text{Im } \alpha \subseteq \{\pm 1, \pm 2\}$ . We know that  $1, 2 \in \text{Im } \alpha$ .

$$-1 : N^2 = -M^4 + 10M^2 e^2 - 8e^4$$

$(1, 1, 1)$  is a solution so  $-1 \in \text{Im } \alpha$ . Thus,  $\text{Im } \alpha = \{\pm 1, \pm 2\}$ . Now, consider  $\text{Im } \bar{\alpha}$ .

$$\bar{E} : y^2 = x^3 - 20x^2 + 68x$$

$$\text{Im } \bar{\alpha} \subseteq \{\pm 1, \pm 2, \pm 17, \pm 34\}$$

By signs,  $\text{Im } \bar{\alpha} \subseteq \{1, 2, 7, 34\}$ . Remains to consider 2, 34:

$$2 : N^2 = 2M^2 - 20M^2e^2 + 34e^2$$

Note that  $(N, M, e) = (4, 1, 1)$  is a solution. Thus,  $\text{Im } \bar{\alpha} = \{1, 2, 7, 34\}$ . By rank formula

$$2^r = \frac{4 \cdot 4}{4} = 4$$

so  $r = 2$ .